

REALTEK

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RTL8211F-CG RTL8211FD-CG
RTL8211FI-CG RTL8211FDI-CG

DRAFT

INTEGRATED 10/100/1000M ETHERNET TRANSCEIVER

DATASHEET

(CONFIDENTIAL: Development Partners Only)

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USING THIS DOCUMENT

This document is intended for the software engineer’s reference and provides detailed programming information.

Though every effort has been made to ensure that this document is current and accurate, more information may have become available subsequent to the production of this guide.

REVISION HISTORY

Revision	Release Date	Summary
1.0	2013/05/20	First release.
1.1	2014/02/21	Added RTL8211FD-CG, RTL8211FI-CG, and RTL8211FDI-CG data.

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1. General Description

The Realtek RTL8211F-CG/RTL8211FD-CG/RTL8211FI-CG/RTL8211FDI-CG is a highly integrated Ethernet transceiver that complies with 10Base-T, 100Base-TX, and 1000Base-T IEEE 802.3 standards. It provides all the necessary physical layer functions to transmit and receive Ethernet packets over CAT.5 UTP cable. The RTL8211FI and RTL8211FDI are manufactured to industrial grade standards.

The RTL8211F(I)/RTL8211FD(I) uses state-of-the-art DSP technology and an Analog Front End (AFE) to enable high-speed data transmission and reception over UTP cable. Functions such as Crossover Detection & Auto-Correction, polarity correction, adaptive equalization, cross-talk cancellation, echo cancellation, timing recovery, and error correction are implemented in the RTL8211F(I)/RTL8211FD(I) to provide robust transmission and reception capabilities at 10Mbps, 100Mbps, or 1000Mbps.

Data transfer between MAC and PHY is via the Reduced Gigabit Media Independent Interface (RGMII) for 1000Base-T, 10Base-T, and 100Base-TX. The RTL8211F(I)/RTL8211FD(I) supports various RGMII signaling voltages, including 3.3, 2.5, 1.8, and 1.5V.

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2. Features

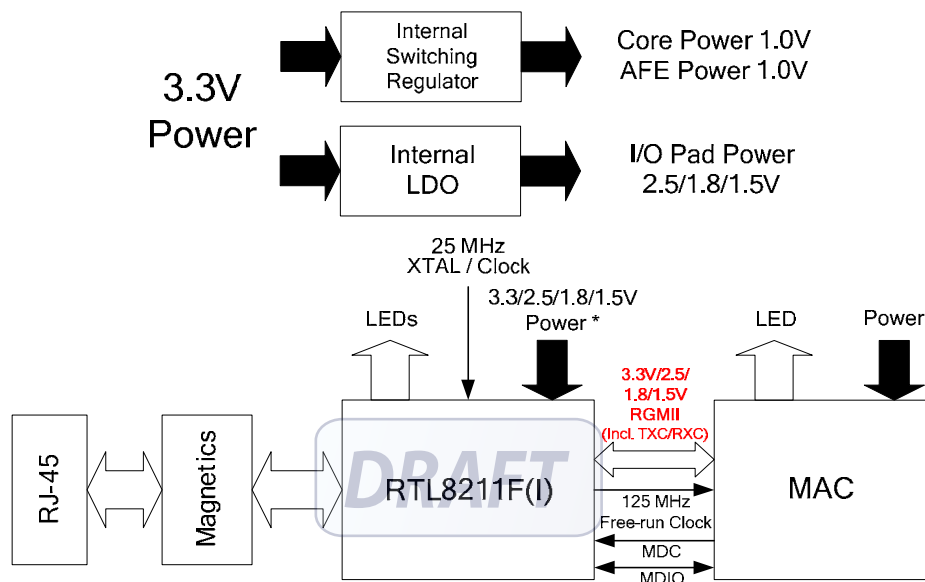
- 1000Base-T IEEE 802.3ab Compliant
- 100Base-TX IEEE 802.3u Compliant
- 10Base-T IEEE 802.3 Compliant
- Supports RGMII
- Supports IEEE 802.3az-2010 (Energy Efficient Ethernet)
- Built-in Wake-on-LAN (WOL)
- Supports Interrupt function
- Supports Parallel Detection
- Crossover Detection & Auto-Correction
- Automatic polarity correction
- Supports PHYRSTB core power Turn-Off
- Baseline Wander Correction
- Supports 120m for CAT.5 cable in 1000Base-T
- Selectable 3.3/2.5/1.8/1.5V signaling for RGMII
- Supports 25MHz external crystal or OSC
- Provides 125MHz clock source for MAC
- Provides 3 network status LEDs
- Supports Link Down power saving
- Green Ethernet (1000/100Mbps mode only)
- Built-in Switching Regulator and LDO
- 40-pin QFN Green Package
- 55 nm process with ultra-low power consumption
- Industrial grade manufacturing process (RTL8211FI/RTL8211FDI)

3. System Applications

- DTV (Digital TV)
- MAU (Media Access Unit)
- CNR (Communication and Network Riser)
- Game Console
- Printer and Office Machine
- DVD Player and Recorder
- Ethernet Hub
- Ethernet Switch

In addition, the RTL8211F(I)/RTL8211FD(I) can be used in any embedded system with an Ethernet MAC that needs a UTP physical connection.

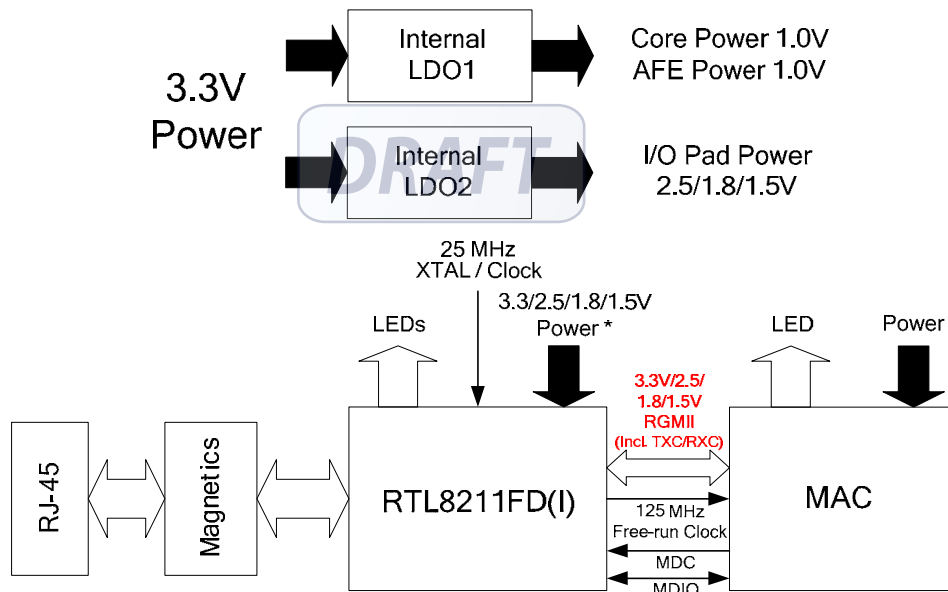
3.1. Application Diagram - RTL8211F(I)



**Note: 3.3/2.5/1.8/1.5V power here means I/O pad power sourced from external power, not from the internal LDO.*

Figure 1. Application Diagram – RTL8211F(I)

3.2. Application Diagram - RTL8211FD(I)



**Note: 3.3/2.5/1.8/1.5V power here means I/O pad power sourced from external power, not from the internal LDO.*

Figure 2. Application Diagram - RTL8211FD(I)

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4. Block Diagram

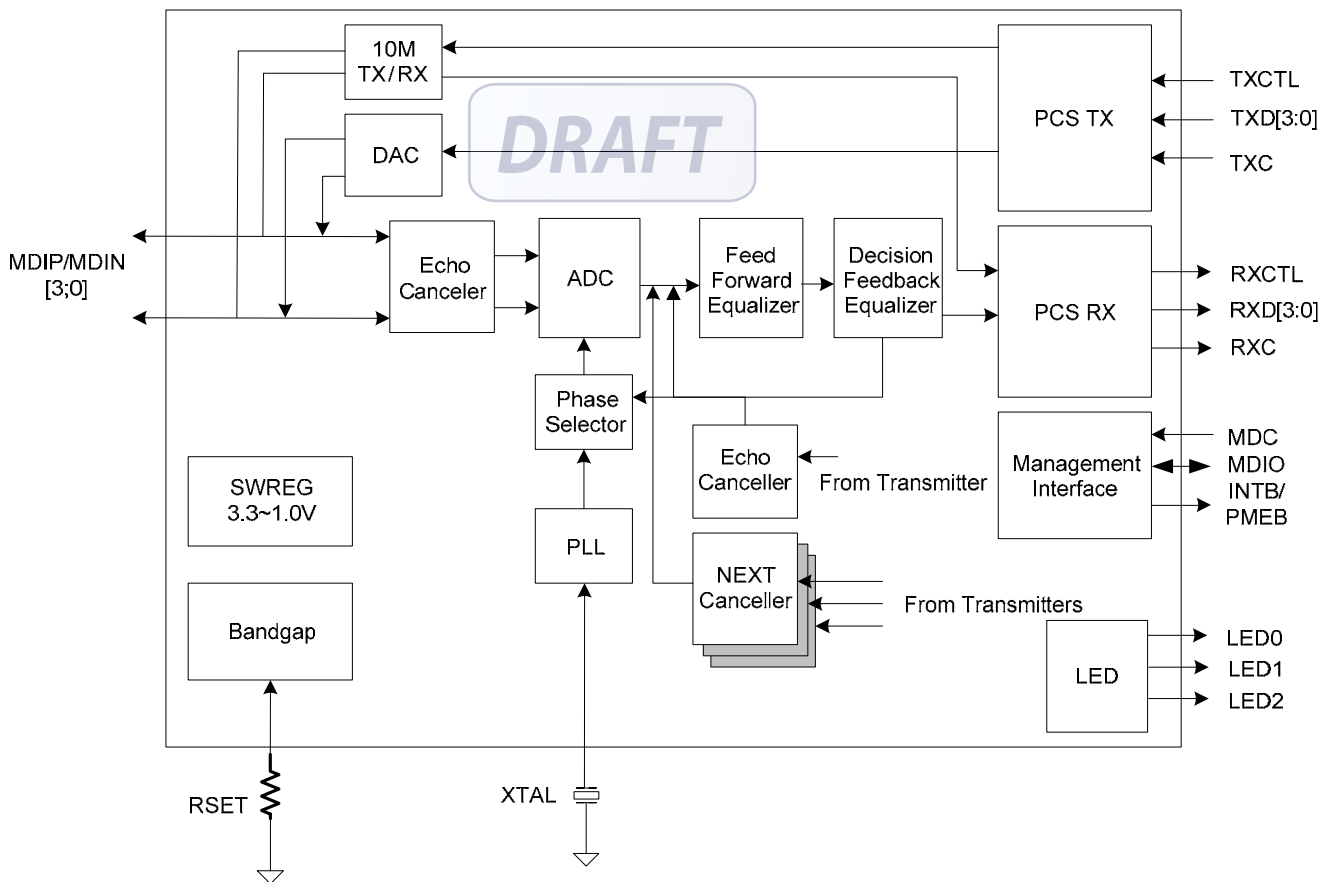


Figure 3. Block Diagram

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5. Pin Assignments

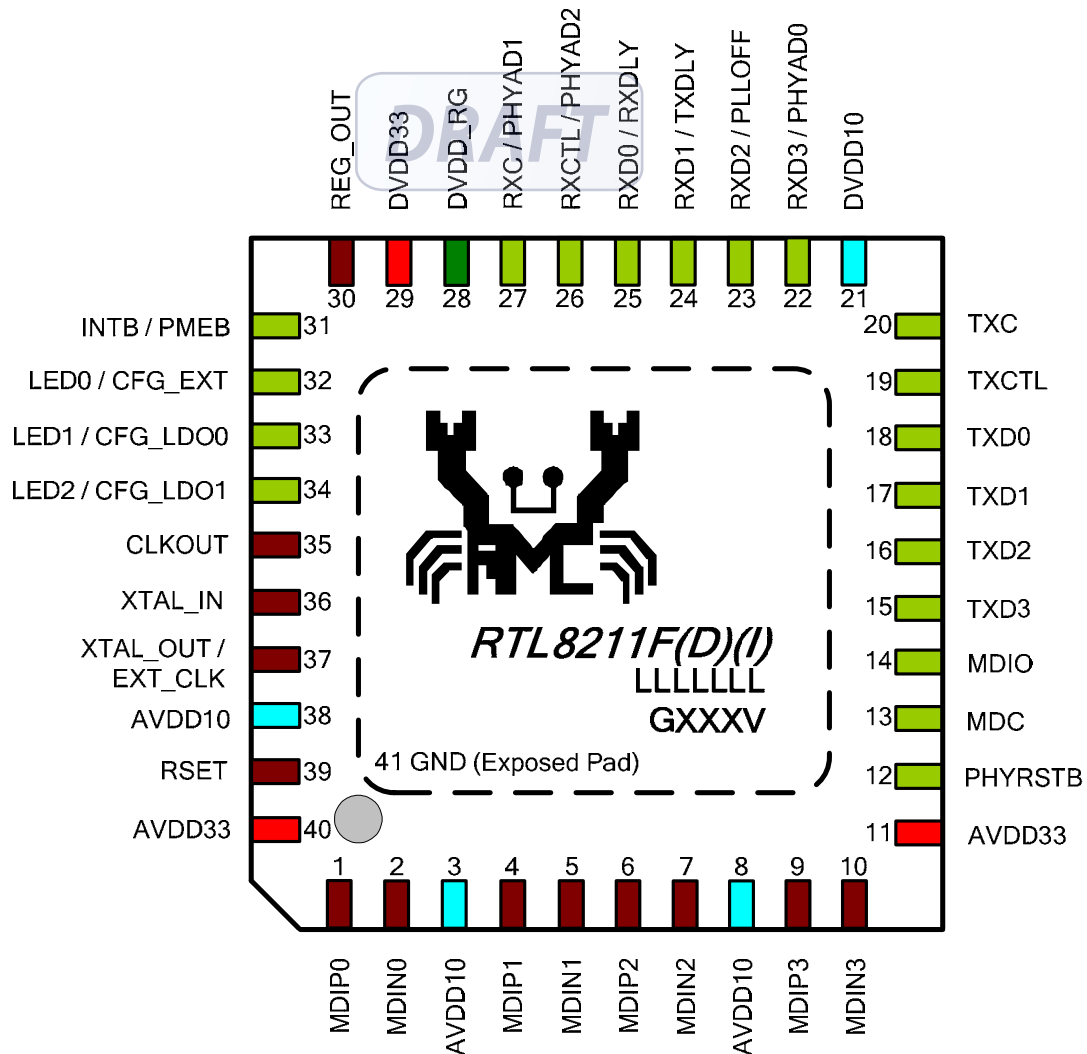


Figure 4. Pin Assignments (40-Pin QFN)

5.1. Package Identification

Green package is indicated by the 'G' in GXXXV (Figure 4).

6. Pin Descriptions

Some pins have multiple functions. Refer to the Pin Assignments figure on page 6 for a graphical representation.

I: Input	LI: Latched Input During Power up or Reset
O: Output	IO: Bi-Directional Input and Output
P: Power	PD: Internal Pull Down During Power On Reset
PU: Internal Pull Up During Power On Reset	OD: Open Drain
G: Ground	

6.1. Transceiver Interface

Table 1. Transceiver Interface

Pin No.	Pin Name	Type	Description
1	MDIP0	IO	In MDI mode, this is the first pair in 1000Base-T, i.e., the BI_DA+/- pair, and is the transmit pair in 10Base-T and 100Base-TX.
2	MDIN0	IO	In MDI crossover mode, this pair acts as the BI_DB+/- pair, and is the receive pair in 10Base-T and 100Base-TX.
4	MDIP1	IO	In MDI mode, this is the second pair in 1000Base-T, i.e., the BI_DB+/- pair, and is the receive pair in 10Base-T and 100Base-TX.
5	MDIN1	IO	In MDI crossover mode, this pair acts as the BI_DA+/- pair, and is the transmit pair in 10Base-T and 100Base-TX.
6	MDIP2	IO	In MDI mode, this is the third pair in 1000Base-T, i.e., the BI_DC+/- pair.
7	MDIN2	IO	In MDI crossover mode, this pair acts as the BI_DD+/- pair.
9	MDIP3	IO	In MDI mode, this is the fourth pair in 1000Base-T, i.e., the BI_DD+/- pair.
10	MDIN3	IO	In MDI crossover mode, this pair acts as the BI_DC+/- pair.

6.2. Clock

Table 2. Clock

Pin No.	Pin Name	Type	Description
36	XTAL_IN	I	25MHz Crystal Input. Connect to GND if an external 25MHz oscillator drives XTAL_OUT/EXT_CLK.
37	XTAL_OUT/E XT_CLK	O	25MHz Crystal Output. If a 25MHz oscillator is used, connect XTAL_OUT/EXT_CLK to the oscillator's output (see section 10.3, page 58 for clock source specifications).
35	CLKOUT	O	125/25MHz Reference Clock Generated from Internal PLL. This pin should be kept floating if this clock is not used by MAC.

6.3. RGMII

Table 3. RGMII

Pin No.	Pin Name	Type	Description
20	TXC	I	The transmit reference clock will be 125MHz, 25MHz, or 2.5MHz depending on speed.
18	TXD0	I	Transmit Data. Data is transmitted from MAC to PHY via TXD[3:0].
17	TXD1	I	
16	TXD2	I	
15	TXD3	I	
19	TXCTL	I	Receive Control Signal from the MAC.
27	RXC	O/LI/PD	The continuous receive reference clock will be 125MHz, 25MHz, or 2.5MHz, and is derived from the received data stream.
25	RXD0	O/LI/PU	Receive Data. Data is transmitted from PHY to MAC via RXD[3:0].
24	RXD1	O/LI/PD	
23	RXD2	O/LI/PD	
22	RXD3	O/LI/PU	
26	RXCTL	O/LI/PD	Transmit Control Signal to the MAC.

6.4. Management Interface

Table 4. Management Interface

Pin No.	Pin Name	Type	Description
13	MDC	I	Management Data Clock.
14	MDIO	IO/PU	Input/Output of Management Data. Pull up 3.3/2.5/1.8/1.5V for 3.3/2.5/1.8/1.5V RGMII, respectively.
31	INTB/PMEB	O/OD	1. Interrupt (supports 3.3V pull up). Set low if status changed; active low. 2. Power Management Event (supports 3.3V pull up). Set low if received a magic packet or wake up frame; active low. Keep this pin floating if either of the functions is not used. <i>Note 1: The behavior of this pin is level-triggered.</i> <i>Note 2: The function of this pin (INTB/PMEB) can be assigned by Page 0xd40, Reg.22, bit[5]:</i> <i>1: Pin 31 functions as PMEB.</i> <i>0: Pin 31 functions as INTB (default)</i>

6.5. Reset

Table 5. Reset

Pin No.	Pin Name	Type	Description
12	PHYRSTB	I	Hardware Reset. Active low. For a complete PHY reset, this pin must be asserted low for at least 10ms. All registers will be cleared after a hardware reset.

6.6. Mode Selection

Table 6. Mode Selection

Pin No.	Pin Name	Type	Description
22	PHYAD0	O/LI/PU	PHYAD[2:0]. PHY Address Configuration.
27	PHYAD1	O/LI/PD	
26	PHYAD2	O/LI/PD	
23	PLLOFF	O/LI/PD	ALDPS Mode PLL Off Configuration. Pull up to stop PLL when entering ALDPS mode. <i>Note:</i> <i>Whole system power consumption in typical ALDPS mode is 39.6mW for the RTL8211F(I), and 69.3mW for the RTL8211FD(I).</i> <i>Whole system power consumption in ALDPS low power mode (with PLL turned off) is 10.3mW for the RTL8211F(I), and 23.1 mW for the RTL8211FD(I).</i> <i>All with 1.8V RGMII.</i>
24	TXDLY	O/LI/PD	RGMII Transmit Clock Timing Control. Pull up to add 2ns delay to TXC for TXD latching
25	RXDLY	O/LI/PU	RGMII Receiver Clock Timing Control. Pull up to add 2ns delay to RXC for RXD latching
32	CFG_EXT	O/LI/PD	IO Pad External Power Source Mode Configuration. Pull up to use the external power source for the IO pad. Pull down to use the integrated LDO to transform the desired voltage for the IO pad.
33	CFG_LDO0	O/LI/PU	LDO Output Voltage Selection for I/O Pad/ External Power Source Voltage Selection for I/O Pad. When pulling down CFG_EXT pin, CFG_LDO[1:0] represent LDO output voltage setting for IO pad: 2'b00: Reserved. 2'b01: 2.5V. 2'b10: 1.8V. 2'b11: 1.5V. When pulling up CFG_EXT pin, CFG_LDO[1:0] stand for input voltage selection of the external power for IO pad: 2'b00: 3.3V. 2'b01: 2.5V. 2'b10: 1.8V. 2'b11: 1.5V.
34	CFG_LDO1	O/LI/PD	

6.7. LED Default Settings

Table 7. LED Default Settings

Pin No.	Pin Name	Type	Description
32	LED0	O/LI/PD	High=Link Up at 10Mbps Blinking=Transmitting or Receiving.
33	LED1	O/LI/PU	Low=Link Up at 100Mbps Blinking=Transmitting or Receiving.
34	LED2	O/LI/PD	High=Link Up at 1000Mbps Blinking=Transmitting or Receiving.

Note: High/Low active depends on hardware configuration setting. See section 7.14 LED Configuration, page 26 for details.

6.8. Regulator and Reference

Table 8. Regulator and Reference

Pin No.	Pin Name	Type	Description
39	RSET	O	Reference. External Resistor Reference.
30	REG_OUT	O	For RTL8211F(I): Switching Regulator 1.0V Output. Connect to a 2.2μH or 4.7μH inductor. For RTL8211FD(I): Low-dropout Regulator 1.0V Output.

6.9. Power and Ground

Table 9. Power and Ground

Pin No.	Pin Name	Type	Description
29	DVDD33	P	Digital Power. 3.3V.
28	DVDD_RG	P	Digital I/O Pad Power. When pulling high CFG_EXT pin during Hardware Configuration (External Power mode), connect this pin to the external power source for 3.3/2.5/1.8/1.5V RGMII I/O.
21	DVDD10	P	Digital Core Power. 1.0V.
11, 40	AVDD33	P	Analog Power. 3.3V.
3, 8, 38	AVDD10	P	Analog Power. 1.0V.
41	GND	G	Ground. Exposed Pad (E-Pad) is Analog and Digital Ground (see section 11 Mechanical Dimensions, page 64).

7. Function Description

7.1. Transmitter

7.1.1. 1000Mbps Mode

The RTL8211F(I)/RTL8211FD(I)'s PCS layer receives data bytes from the MAC through the RGMII interface and performs generation of continuous code-groups through 4D-PAM5 coding technology. These code groups are passed through a waveform-shaping filter to minimize EMI effect, and are transmitted onto the 4-pair CAT.5 cable at 125MBaud/s through a D/A converter.

7.1.2. 100Mbps Mode

The transmitted 4-bit nibbles (TXD[3:0]) from the MAC, clocked at 25MHz (TXCLK), are converted into 5B symbol code through 4B/5B coding technology, then through scrambling and serializing, are converted to 125MHz NRZ and NRZI signals. The NRZI signals are passed to the MLT3 encoder, then to the D/A converter and transmitted onto the media.

7.1.3. 10Mbps Mode

The transmit 4-bit nibbles (TXD[3:0]) from the MAC, clocked at 2.5MHz (TXCLK), are serialized into 10Mbps serial data. The 10Mbps serial data is converted into a Manchester-encoded data stream and is transmitted onto the media by the D/A converter.

7.2. Receiver

7.2.1. 1000Mbps Mode

Input signals from the media first pass through the on-chip sophisticated hybrid circuit to subtract the transmitted signal from the input signal for effective reduction of near-end echo. The received signal is processed with state-of-the-art technology, such as adaptive equalization, BLW (Baseline Wander) correction, cross-talk cancellation, echo cancellation, timing recovery, error correction, and 4D-PAM5 decoding. The 8-bit-wide data is recovered and is sent to the RGMII interface at a clock speed of 125MHz. The Rx MAC retrieves the packet data from the receive RGMII interface and sends it to the Rx Buffer Manager.

7.2.2. 100Mbps Mode

The MLT3 signal is processed with an ADC, equalizer, BLW (Baseline Wander) correction, timing recovery, MLT3 and NRZI decoder, descrambler, 4B/5B decoder, and is then presented to the RGMII interface in 4-bit-wide nibbles at a clock speed of 25MHz.

7.2.3. 10Mbps Mode

The received differential signal is converted into a Manchester-encoded stream first. Next, the stream is processed with a Manchester decoder, and is de-serialized into 4-bit-wide nibbles. The 4-bit nibbles are presented to the RGMII interface at a clock speed of 2.5MHz.

7.3. *Energy Efficient Ethernet (EEE)*

The RTL8211F(I)/RTL8211FD(I) supports IEEE 802.3az-2010, also known as Energy Efficient Ethernet (EEE), at 10Mbps, 100Mbps, and 1000Mbps. It provides a protocol to coordinate transitions to/from a lower power consumption level (Low Power Idle mode) based on link utilization. When no packets are being transmitted, the system goes to Low Power Idle mode to save power. Once packets need to be transmitted, the system returns to normal mode, and does this without changing the link status and without dropping/corrupting frames.

To save power, when the system is in Low Power Idle mode, most of the circuits are disabled, however, the transition time to/from Low Power Idle mode is kept small enough to be transparent to upper layer protocols and applications.

EEE also specifies a negotiation method to enable link partners to determine whether EEE is supported.

Refer to <http://www.ieee802.org/3/az/index.html> for more details.

7.4. *Wake-On-LAN (WOL)*

The RTL8211F(I)/RTL8211FD(I) can monitor the network for a Wakeup Frame or a Magic Packet, and notify the system via the PMEB (Power Management Event; 'B' means low active) pin when such a packet or event occurs¹. The system can then be restored to a normal state to process incoming jobs. The PMEB pin needs to be connected with a 4.7k-ohm resistor and pulled up to 3.3V. When the Wakeup Frame or a Magic Packet is sent to the PHY, the PMEB pin will be set low to notify the system to wake up. Refer to the WOL application note for details.

Magic Packet Wakeup occurs only when the following conditions are met:

- The destination address of the received Magic Packet is acceptable to the RTL8211F(I)/RTL8211FD(I), e.g., a broadcast, multicast, or unicast packet addressed to the current RTL8211F(I)/RTL8211FD(I).
- The received Magic Packet does not contain a CRC error.
- The Magic Packet pattern matches; i.e., 6 * FFh + MISC (can be none) + 16 * DID (Destination ID) in any part of a valid Ethernet packet.

A Wakeup Frame event occurs only when the following conditions are met:

- The destination address of the received Wakeup Frame is acceptable to the RTL8211F(I)/RTL8211FD(I), e.g., a broadcast, multicast, or unicast address to the current RTL8211F(I)/RTL8211FD(I).
- The received Wakeup Frame does not contain a CRC error.
- The 16-bit CRC² of the received Wakeup Frame matches the 16-bit CRC of the sample Wakeup Frame pattern given by the local machine's OS. Or, the RTL8211F(I)/RTL8211FD(I) is configured to allow direct packet wakeup, e.g., a broadcast, multicast, or unicast network packet.

Note 1: The INTB and PMEB functions share the same pin (pin 31), and can be determined by Page 0xd40, Reg.22, bit[5].

Note 2: 16-bit CRC: The RTL8211F(I)/RTL8211FD(I) supports eight long wakeup frames (covering 128 mask bytes from offset 0 to 127 of any incoming network packet). CRC16 polynomial= $x^{16}+x^{12}+x^5+1$.

7.5. Interrupt

Whenever there is a status change on the media detected by the RTL8211F(I)/RTL8211FD(I), they will drive the interrupt pin (INTB) low to issue an interrupt event. The MAC senses the status change and accesses the registers (Page 0, Register 19) through the MDC/MDIO interface in response.

Once these status registers (Page 0, Register 19) have been read by the MAC through the MDC/MDIO, the INTB is de-asserted. The RTL8211F(I)/RTL8211FD(I) interrupt function removes the need for continuous polling through the MDC/MDIO management interface.

Note 1: The interrupt of the RTL8211F(I)/RTL8211FD(I) is a level-triggered mechanism.

Note 2: The INTB and PMEB functions share the same pin (pin 31), and can be determined by Page 0xd40, Reg.22, bit[5].

7.6. INTB/PMEB Pin Usage

The INTB/PMEB pin (pin 31) of the RTL8211F(I)/RTL8211FD(I) is designed to notify in cases of both interrupt and WOL events. The default mode of this pin is INTB (Page 0xd40, Reg.22, bit[5] = 0). For general use, indication of a WOL event is also integrated into one of the interrupt events (Page 0, Reg 18, bit[7] which is triggered when any specified WOL event occurs. However, the 'Low Pulse' waveform format is not supported during this mode; only the active low, level-triggered waveform is provided.

If PMEB mode is selected (Page 0xd40, Reg.22, bit[5] = 1), pin 31 becomes a fully functional PMEB pin. Note that the interrupt function is disabled in this mode.

7.7. MDI Interface

This interface consists of four signal pairs; MDI0, MDI1, MDI2, and MDI3. Each signal pair consists of two bi-directional pins that can transmit and receive at the same time. The MDI interface has internal termination resistors to reduce BOM cost and PCB complexity. For 1000Base-T, all four pairs are used in both directions at the same time. For 10/100 links and during auto-negotiation, only pairs A and B are used.

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7.8. Hardware Configuration

The I/O pad voltage, interface mode, and PHY address can be set by the CONFIG pins. The respective value mapping of CONFIG with the configurable vector is listed in Table 10. To set the CONFIG pins, an external pull-high or pull-low via resistor is required.

Table 10. CONFIG Pins vs. Configuration Register

CONFIG Pin	Configuration
RXD3	PHYAD[0]
RXC	PHYAD[1]
RXCTL	PHYAD[2]
RXD2	PLLOFF
RXD1	TXDLY
RXD0	RXDLY
LED0	CFG_EXT
LED1	CFG_LDO[0]
LED2	CFG_LDO[1]

Table 11. Configuration Register Definitions

Configuration	Description
PHYAD[2:0]	PHY Address. PHYAD sets the PHY address for the device. The RTL8211F(I)/ RTL8211FD(I) supports PHY addresses from 00001 to 00111. <i>Note 1: An MDIO command with PHY address=0 is a broadcast from the MAC; each PHY device should respond. This function can be disabled by setting Reg24.13=0 (See Table 37).</i> <i>Note 2: The RTL8211F(I)/ RTL8211FD(I) with PHYAD[2:0]=000 can automatically remember the first non-zero PHY address. This function can be enabled by setting Reg24.6 = 1 (See Table 37).</i>
PLLOFF	ALDPS Mode PLL Off Configuration. 1: Stop PLL when entering ALDPS mode (via 4.7k-ohm to DVDD_RG) 0: PLL continue toggling when entering ALDPS mode (via 4.7k-ohm to GND)
TXDLY	RGMII Transmit Clock Timing Control. 1: Add 2ns delay to RXC for RXD latching (via 4.7k-ohm to DVDD_RG) 0: No delay (via 4.7k-ohm to GND)
RXDLY	RGMII Receive Clock Timing Control. 1: Add 2ns delay to RXC for RXD latching (via 4.7k-ohm to DVDD_RG) 0: No delay (via 4.7k-ohm to GND)
CFG_EXT	I/O Pad External Power Source Mode Configuration. 1: Use the external power source for the IO pad (via 4.7k-ohm to 3.3V) 0: Use the integrated LDO to transform the desired voltage for the IO pad (via 4.7k-ohm to GND)

Configuration	Description
CFG_LDO[1:0]	LDO Output Voltage Selection for I/O pad/ External Power Source Voltage Selection for I/O pad. When pulling down CFG_EXT pin, CFG_LDO[1:0] represent LDO output voltage setting for IO pad: (via 4.7k-ohm to GND) 00: Reserved. 01: 2.5V 10: 1.8V 11: 1.5V When pulling up CFG_EXT pin, CFG_LDO[1:0] stand for external power voltage selection for IO pad: (via 4.7k-ohm to 3.3V) 00: 3.3V 01: 2.5V 10: 1.8V 11: 1.5V

7.9. LED and PHY Address Configuration

In order to reduce the pin count on the RTL8211F(I)/RTL8211FD(I), the LED pins are duplexed with the CFG_EXT and CFG_LDO pins. As the Hardware Configuration shares the LED output pins, the external combinations required for strapping and LED usage must be considered in order to avoid contention. Specifically, when the LED outputs are used to drive LEDs directly, the active state of each output driver is dependent on the logic level sampled by the corresponding CFG_EXT/CFG_LDO inputs upon power-on/reset.

For example, as Figure 5 (left-side) shows, if a given CFG_EXT/CFG_LDO inputs are resistively pulled high then the corresponding LED outputs will be configured as an active low driver. On the right side, we can see that if a given CFG_EXT/CFG_LDO inputs are resistively pulled low then the corresponding output will be configured as an active high driver. The Hardware Configuration pins should not be connected to GND or VCC directly, but must be pulled high or low through a resistor (e.g., 4.7KΩ). If no LED indications are needed, the components of the LED path (LED+510Ω) can be removed.

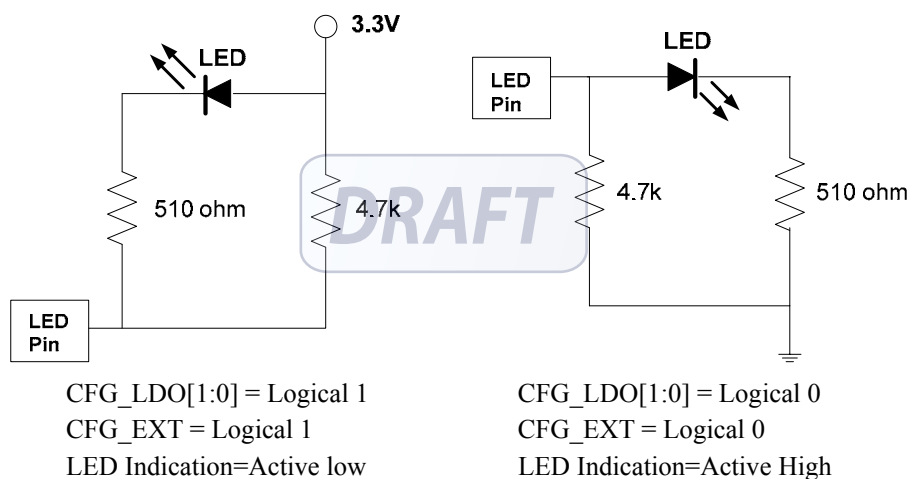


Figure 5. LED and PHY Address Configuration

7.10. Green Ethernet (1000/100Mbps Mode Only)

7.10.1. Cable Length Power Saving

In 1000/100Mbps mode the RTL8211F(I)/RTL8211FD(I) provides dynamic detection of cable length and dynamic adjustment of power required for the detected cable length. This feature provides intermediate performance with minimum power consumption.

7.10.2. Register Setting

Follow the register settings below to DISABLE Green Ethernet (Default is ‘Enabled’)

Write Reg27, Data=0x8011

Write Reg28, Data=0x573f

Follow the register settings below to ENABLE Green Ethernet (Default is ‘Enabled’)

Write Reg27, Data=0x8011

Write Reg16, Data=0xd73f

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7.11. MAC/PHY Interface

The RTL8211F(I)/RTL8211FD(I) supports industry standards and is suitable for most off-the-shelf MACs with an RGMII interface.

7.11.1. RGMII

Among the RGMII interface in 100Base-TX and 10Base-T modes, TXC and RXC sources are 25MHz and 2.5MHz respectively; while in 1000Base-T mode, TXC and RXC sources are 125MHz. TXC will always be generated by the MAC and RXC will always be generated by the PHY. TXD[3:0] and RXD[3:0] signals are used for data transitions on the rising and falling edge of the clock.

7.11.2. Management Interface

The management interface provides access to the internal registers through the MDC and MDIO pins as described in IEEE 802.3u section 22. The MDC signal, provided by the MAC, is the management data clock reference to the MDIO signal. The MDIO is the management data input/output and is a bi-directional signal that runs synchronously to MDC. The MDIO pin needs a 1.5k Ohm pull-up resistor to maintain the MDIO high during idle and turnaround.

The RTL8211F(I)/RTL8211FD(I) can share the same MDIO line. In switch/router applications, each port should be assigned a unique address during the hardware reset sequence, and it can only be addressed via that unique PHY address. For detailed information on the management registers, see section 8 Register Descriptions, page 29.

Table 12. Management Frame Format

	Management Frame Fields							
	Preamble	ST	OP	PHYAD	REGAD	TA	DATA	IDLE
Read	1...1	01	10	AAAAA	RRRRR	Z0	DDDDDDDDDDDDDDDDDD	Z
Write	1...1	01	01	AAAAA	RRRRR	10	DDDDDDDDDDDDDDDDDD	Z

Table 13. Management Frame Description

Name	Description
Preamble	32 Contiguous Logical 1's Sent by the MAC on MDIO, along with 32 Corresponding Cycles on MDC. This provides synchronization for the PHY.
ST	Start of Frame. Indicated by a 01 pattern.
OP	Operation Code. Read: 10 Write: 01
PHYAD	PHY Address. Up to eight PHYs can be connected to one MAC. This 3-bit field selects which PHY the frame is directed to.
REGAD	Register Address. This is a 5-bit field that sets which of the 32 registers of the PHY this operation refers to.

Name	Description
TA	Turnaround. This is a 2-bit-time spacing between the register address and the data field of a frame to avoid contention during a read transaction. For a read transaction, both the STA and the PHY remain in a high-impedance state for the first bit time of the turnaround. The PHY drives a zero bit during the second bit time of the turnaround of a read transaction.
DATA	Data. These are the 16 bits of data.
IDLE	Idle Condition. Not truly part of the management frame. This is a high impedance state. Electrically, the PHY's pull-up resistor will pull the MDIO line to a logical '1'.

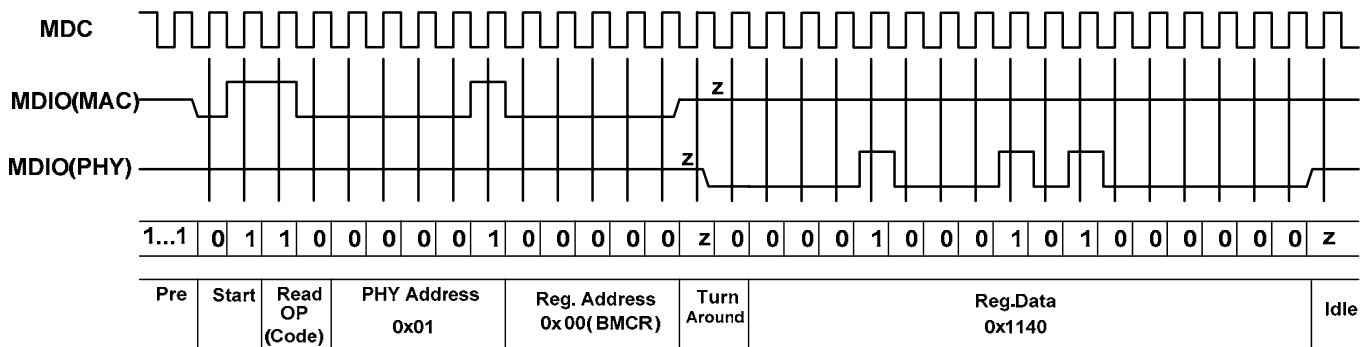


Figure 6. MDC/MDIO Read Timing

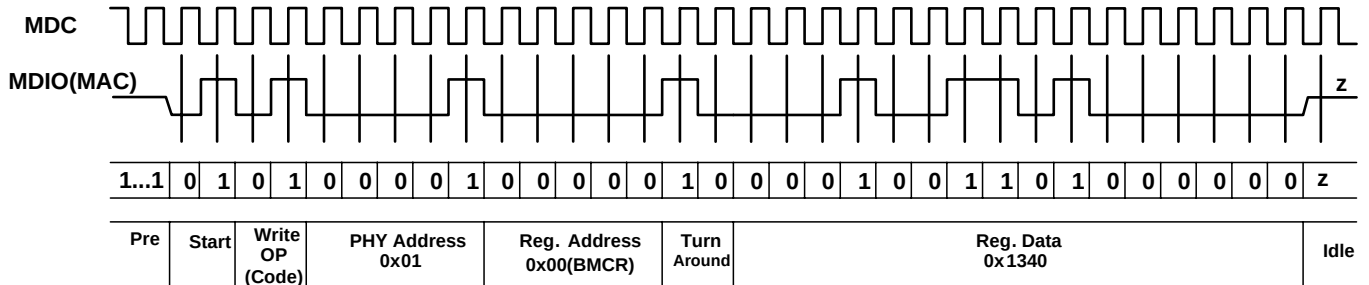


Figure 7. MDC/MDIO Write Timing

7.11.3. Change Page

Set MDIO commands as shown below to switch to the other Page 0xXY (in Hex).

1. Write Register 31 Data=0x00XY (Page XY)
2. Read/Write the target Register Data
3. Write Register 31 Data=0x0000 (switch to Page 0)

7.11.4. Access to MDIO Manageable Device (MMD)

The MDIO Manageable Device (MMD) is an extension to the management interface that provides the ability to access more device registers while still retaining logical compatibility with the MDIO interface, defined in section 8.1 Register Mapping and Definitions, page 29. Access to MMD configuration is provided via Registers 13 and 14.

MMD Read/Write Operation

1. Write Function field to 00 (address mode) and DEVAD field to the device address value for the desired MMD (Register 13).
2. Write the desired address value to the MMD's address register (Register 14).
3. Write Function field to 01 (data mode; no post increment) and DEVAD field to the same device address for the desired MMD (Register 13).
4. Read: Go to step 5. Write: Go to step 6.
5. Read the content of the selected register in MMD (Register 14).
6. Write the content of the selected register in MMD (Register 14).

7.12. Auto-Negotiation

Auto-Negotiation is a mechanism to determine the fastest connection between two link partners. For copper media applications, it was introduced in IEEE 802.3u for Ethernet and Fast Ethernet, and then in IEEE 802.3ab to address extended functions for Gigabit Ethernet. It performs the following:

- Auto-Negotiation Priority Resolution
- Auto-Negotiation Master/Slave Resolution
- Auto-Negotiation PAUSE/ASYMMETRIC PAUSE Resolution
- Crossover Detection & Auto-Correction Resolution

Upon de-assertion of a hardware reset, the RTL8211F(I)/RTL8211FD(I) can be configured to have auto-negotiation enabled, or be set to operate in 10Base-T, 100Base-TX, or 1000Base-T mode via the ANAR and GBCR register (register 4 and 9).

The auto-negotiation process is initiated automatically upon any of the following:

- Power-up
- Hardware reset
- Software reset (register 0.15)

- Restart auto-negotiation (register 0.9)
- Transition from power down to power up (register 0.11)
- Entering the link fail state

Table 14. 1000Base-T Base and Next Page Bit Assignments

Bit	Name	Bit Description	Register Location
Base Page			
D15	NP	Next Page. 1: Indicates that Next Pages follow 0: Indicates that no Next Pages follow	-
D14	Ack	Acknowledge. 1: Indicates that a device has successfully received its link partner's Link Code Word (LCW)	-
D13	RF	Remote Fault. 1: Indicates to its link partner that a device has encountered a fault condition	-
D[12:5]	A[7:0]	Technology Ability Field. Indicates to its link partner the supported technologies specific to the selector field value.	Register 4.[12:5] Table 26, page 33.
D[4:0]	S[4:0]	Selector Field. Always 00001. Indicates to its link partner that it is an IEEE 802.3 device.	Register 4.[4:0] Table 26, page 33.
PAGE 0 (Message Next Page)			
M15	NP	Next Page. 1: Indicates that Next Pages follow 0: Indicates that no Next Pages follow	-
M14	Ack	Acknowledge. 1: Indicates that a device has successfully received its link partner's Link Code Word (LCW)	-
M13	MP	Message Page. 1: Indicates to its link partner that this is a message page, not an unformatted page.	-
M12	Ack2	Acknowledge 2. 1: Indicates to its link partner that the device has the ability to comply with the message.	-
M11	T	Toggle. Used by the NWay arbitration function to ensure synchronization with its link partner during Next Page exchange.	-
M[10:0]	-	1000Base-T Message Code (Always 8).	-
PAGE 1 (Unformatted Next Page)			
U15	NP	Next Page. 1: Indicates that Next Pages follow 0: Indicates that no Next Pages follow	-
U14	Ack	Acknowledge. 1: Indicates that a device has successfully received its link partner's Link Code Word (LCW)	-

Bit	Name	Bit Description	Register Location
U13	MP	Message Page. 1: Indicates to its link partner that this is a message page, not an unformatted page.	-
U12	Ack2	Acknowledge 2. 1: Indicates to its link partner that the device has the ability to comply with the message.	-
U11	T	Toggle. Used by the NWay arbitration function to ensure synchronization with its link partner during Next Page exchange.	-
U[10:5]	-	Reserved. Transmit as 0	-
U4	-	1000Base-T Half Duplex. 1: Half duplex 0: No half duplex	-
U3	-	1000Base-T Full Duplex. 1: Full duplex 0: No full duplex	-
U2	-	1000Base-T Port Type Bit. 1: Multi-port device 0: Single-port device	Register 9.10 (GBCR) Table 31, page 37.
U1	-	1000Base-T Master-Slave Manual Configuration Value. 1: Master 0: Slave This bit is ignored if bit 9.12=0	Register 9.11 (GBCR) Table 31, page 37.
U0	-	1000Base-T Master-Slave Manual Configuration Enable. 1: Manual Configuration Enable This bit is intended to be used for manual selection in Master-Slave mode, and is to be used in conjunction with bit 9.11	Register 9.12 (GBCR) Table 31, page 37.
PAGE 2 (Unformatted Next Page)			
U15	NP	Next Page. 1: Indicates that Next Pages follow 0: Indicates that no Next Pages follow	-
U14	Ack	Acknowledge. 1: Indicates that a device has successfully received its link partner's Link Code Word (LCW)	-
U13	MP	Message Page. 1: Indicates to its link partner that this is a message page, not an unformatted page	-
U12	Ack2	Acknowledge 2. 1: Indicates to its link partner that the device has the ability to comply with the message	-
U11	T	Toggle. Used by the NWay arbitration function to ensure synchronization with its link partner during Next Page exchange.	-
U[10:0]	-	1000Base-T Master-Slave Seed Bit[10:0]	Master-Slave Seed Value SB[10:0]

7.12.1. Auto-Negotiation Priority Resolution

Upon the start of auto-negotiation, to advertise its capabilities each station transmits a 16-bit packet called a Link Code Word (LCW), within a burst of 17 to 33 Fast Link Pulses (FLP). A device capable of auto-negotiation transmits and receives the FLPs. The receiver must identify three identical LCWs before the information is authenticated and used in the arbitration process. The devices decode the base LCW and select capabilities with the highest common denominator supported by both devices.

To advertise 1000Base-T capability, both link partners, sharing the same link medium, should engage in Next Page (1000Base-T Message Page, Unformatted Page 1, and Unformatted Page 2) exchange.

Auto-negotiation ensures that the highest priority protocol will be selected as the link speed based on the following priorities advertised through the Link Code Word (LCW) exchange. Refer to IEEE 802.3 Clause 28 for detailed information.

1. 1000Base-T Full Duplex (highest priority)
2. 1000Base-T Half Duplex
3. 100Base-TX Full Duplex
4. 100Base-TX Half Duplex
5. 10Base-T Full Duplex
6. 10Base-T Half Duplex (lowest priority)

7.12.2. Auto-Negotiation Master/Slave Resolution

To establish a valid 1000Base-T link, the Master/Slave mode of both link partners should be resolved through the auto-negotiation process:

- Master Priority
 - Multi-port > Single-port
 - Manual > Non-manual
- Determination of Master/Slave configuration from LCW
 - Manual_MASTER=U0*U1
 - Manual_SLAVE=U0*!U1
 - Single-port device=!U0*!U2
 - Multi-port device=!U0*U2

Where: U0 is bit 0 of the Unformatted Page 1
U1 is bit 1 of the Unformatted Page 1
U2 is bit 2 of the Unformatted Page 1

- Where there are two stations with the same configuration, the one with higher Master-Slave seed SB[10:0] in the unformatted page 2 shall become Master.
- Master-Slave configuration process resolution:
 - Successful: Bit 10.15 Master-Slave Configuration Fault is set to logical 0, and bit 10.14 is set to logical 1 for Master resolution, or set to logical 0 for Slave resolution.
 - Unsuccessful: Auto-Negotiation restarts.
 - Fault Detect: Bit 10.15 is set to logical 1 to indicate that a configuration fault has been detected. Auto-Negotiation restarts automatically. This happens when both stations are set to manual Master mode or manual Slave mode, or after seven attempts to configure the Master-Slave relationship through the seed method has failed.

7.12.3. Auto-Negotiation PAUSE/ASYMMETRIC PAUSE Resolution

Auto-negotiation is also used to determine the flow control capability between link partners. Flow control is a mechanism that can force a busy transmitting link partner to stop transmitting in a full duplex environment by sending special MAC control frames. In IEEE 802.3u, a PAUSE control frame had already been defined. However, in IEEE 802.3ab, a new ASY-PAUSE control frame was defined; if the MAC can only generate PAUSE frames but is not able to respond to PAUSE frames generated by the link partner, then it is called ASYMMETRIC PAUSE.

PAUSE/ASYMMETRIC PAUSE capability can be configured by setting the ANAR bits 10 and 11 (Table 26, page 33). Link partner PAUSE capabilities can be determined from ANLPAR bits 10 and 11 (Table 27, page 34). A PHY layer device such as the RTL8211F(I)/RTL8211FD(I) is not directly involved in PAUSE resolution, but simply advertises and reports PAUSE capabilities during the Auto-Negotiation process. The MAC is responsible for final PAUSE/ASYMMETRIC PAUSE resolution after a link is established, and is responsible for correct flow control actions thereafter.

7.13. Crossover Detection and Auto-Correction

Ethernet needs a crossover mechanism between both link partners to cross the transmit signal to the receiver when the medium is twisted-pair cable. Crossover Detection & Auto-Correction Configuration eliminates the need for crossover cables between devices, such as two computers connected to each other with a CAT.5 Ethernet cable. The basic concept is to assume the initial default setting is MDI mode, and then check the link status. If no link is established after a certain time, change to MDI Crossover mode and repeat the process until a link is established. An 11-bit pseudo-random timer is applied to decide the mode change time interval.

Crossover Detection & Auto-Correction is not a part of the Auto-Negotiation process, but it utilizes the process to exchange the MDI/MDI Crossover configuration. If the RTL8211F(I)/RTL8211FD(I) is configured to only operate in 100Base-TX or only in 10Base-T mode, then Auto-Negotiation is disabled only if the Crossover Detection & Auto-Correction function is also disabled. If Crossover Detection & Auto-Correction are enabled, then Auto-Negotiation is enabled and the RTL8211F(I)/RTL8211FD(I) advertises only 100Base-TX mode or 10Base-T mode. If the speed of operation is configured manually and Auto-Negotiation is still enabled because the Crossover Detection & Auto-Correction function is enabled, then the duplex advertised is as follows:

1. If it is set to half duplex, then only half duplex is advertised.
2. If it is set to full duplex, then both full and half duplex are advertised.

If the user wishes to advertise only full duplex at a particular speed with the Crossover Detection & Auto-Correction function enabled, then Auto-Negotiation should be enabled (register 0.12) with the appropriate advertising capabilities set in registers 4 or 9. The Crossover Detection & Auto-Correction function may be enabled/disable by setting (register 24.[9:8]) manually.

After initial configuration following a hardware reset, Auto-Negotiation can be enabled and disabled via register 0.12, speed via registers 0.13, 0.6, and duplex via register 0.8. The abilities that are advertised can be changed via registers 4 and 9. Changes to registers 0.12, 0.13, 0.6, and 0.8 do not take effect unless at least one of the following events occurs:

- Software reset (register 0.15)
- Restart of Auto-Negotiation (register 0.9)
- Transition from power-down to power-up (register 0.11)

Registers 4 and 9 are internally latched once each time Auto-Negotiation enters the ABILITY DETECT state in the arbitration state machine (IEEE 802.3). Hence a write into register 4 or 9 has no effect once the RTL8211F(I)/RTL8211FD(I) begins to transmit Fast Link Pulses.

Register 7 is treated in a similar manner as 4 and 9 during additional Next Page exchanges. Once the RTL8211F(I)/RTL8211FD(I) completes Auto-Negotiation, it updates the various statuses in registers 1, 5, 6, and 10. The speed, duplex, page received, and Auto-Negotiation completed statuses are also available in registers 26 and 29.

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7.14. LED Configuration

7.14.1. Customized LED Function

The RTL8211F(I)/RTL8211FD(I) supports three LED pins, suitable for multiple types of applications that can directly drive the LEDs. The output of these pins is determined by setting the corresponding bits in Page 0xd04 Register 16. The functionality of the RTL8211F(I)/RTL8211FD(I) LEDs is shown in Table 15.

Table 15. LED Default Definitions

Pin	Description
LED0	10M Link and Active (Transmitting or Receiving)
LED1	100M Link and Active (Transmitting or Receiving)
LED2	1000M Link and Active (Transmitting or Receiving)

The LED pins can be customized from Page 0xd04 Register 16. To change the register page, see note (below) and Table 16 LED Register Table. There are 16 configuration types (see Table 17 LED Configuration Table, page 27).

Note: To switch to Page 0xd04, set Register 31 Data=0x0d04 (set page). After LED setting, switch to PHY's Page 0 (Register 31 Data=0x0000).

Table 16. LED Register Table

	LINK Speed			Active (Tx/Rx)
	10Mbps	100Mbps	1000Mbps	
LED0	Reg16 Bit0	Reg16 Bit1	Reg16 Bit3	Reg16 Bit4
LED1	Reg16 Bit5	Reg16 Bit6	Reg16 Bit8	Reg16 Bit9
LED2	Reg16 Bit10	Reg16 Bit11	Reg16 Bit13	Reg16 Bit14

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Table 17. LED Configuration Table

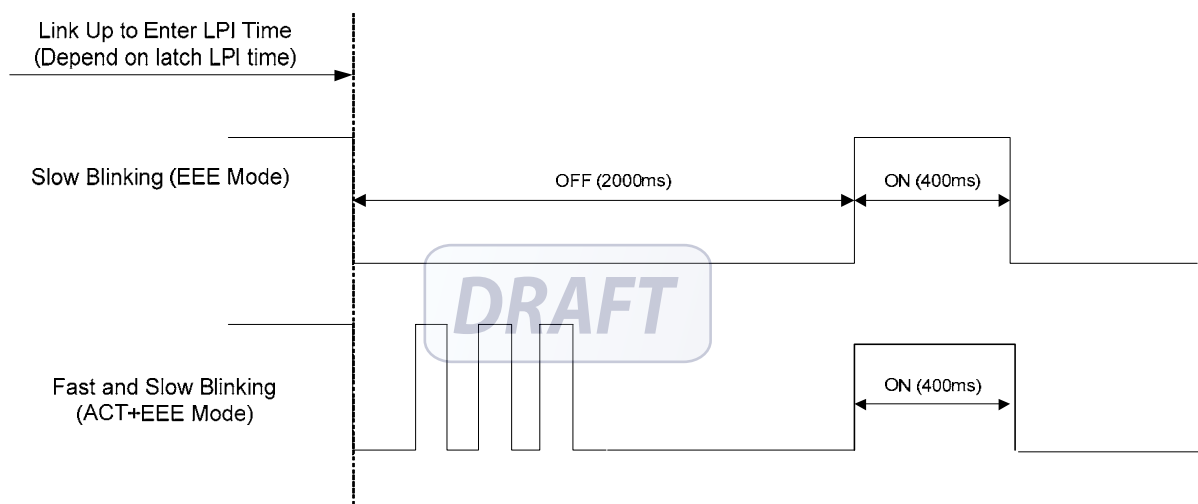
Pin	LINK Bit			Active (TX/RX) Bit	Description
	10Mbps	100 Mbps	1000 Mbps		
LED	0	0	0	0	N/A
	0	0	0	1	N/A
	0	0	1	0	Link 1000
	0	0	1	1	Link 1000+Active
	0	1	0	0	Link 100
	0	1	0	1	Link 100+Active
	0	1	1	0	Link 100/1000
	0	1	1	1	Link 100/1000+Active
	1	0	0	0	Link 10
	1	0	0	1	Link 10+Active
	1	0	1	0	Link 10/1000
	1	0	1	1	Link 10/1000+Active
	1	1	0	0	Link 10/100
	1	1	0	1	Link 10/100+Active
	1	1	1	0	Link 10/100/1000
	1	1	1	1	Link 10/100/1000+Active

7.14.2. EEE LED Function

EEE Idle mode: LED continuous slow blinking.

EEE Active mode: LED fast and slow blinking (on packet transmitting and receiving).

Refer to Table 49, page 45 for EEE LED enable setting.


Figure 8. EEE LED Behavior

7.15. Polarity Correction

The RTL8211F(I)/RTL8211FD(I) automatically corrects polarity errors on the receive pairs in 1000Base-T and 10Base-T modes. In 100Base-TX mode polarity is irrelevant. In 1000Base-T mode, receive polarity errors are automatically corrected based on the sequence of idle symbols. Once the descrambler is locked, the polarity is also locked on all pairs. The polarity becomes unlocked only when the receiver loses lock.

In 10Base-T mode, polarity errors are corrected based on the detection of validly spaced link pulses. The detection begins during the MDI crossover detection phase and locks when the 10Base-T link is up. The polarity becomes unlocked when the link is down.

7.16. Power

A voltage regulator is implemented to generate operating power (switching regulator for the RTL8211F(I); LDO for the RTL8211FD(I)). The system vendor needs to supply a 3.3V, 1A steady power source. The RTL8211F(I)/RTL8211FD(I) converts the 3.3V steady power source to 1.0V via a switching regulator/LDO.

Another possible implementation is to use an external regulator to generate 1.0V. Be sure that the regulator meets the required current rate (0.95V~1.05V).

The RTL8211F(I)/RTL8211FD(I) implements an option for the RGMII power pins. The standard I/O voltage of the RGMII interface is 3.3V, with support for 2.5/1.8/1.5V to lower EMI. The 2.5/1.8/1.5V power source for RGMII is supplied from an internal LDO or from an external power source.

7.17. PHY Reset (Hardware Reset)

The RTL8211F(I)/RTL8211FD(I) has a PHYRSTB pin to reset the chip. For a complete PHY reset, this pin must be asserted low for at least 10ms (T_{gap} in Figure 9) for the internal regulator. Wait for a further 30ms (for internal circuits settling time) before accessing the PHY register. All registers will return to default values after a hardware reset. Refer to the RTL8211xx-xx_Power_Sequence_App_Note for more detailed information.

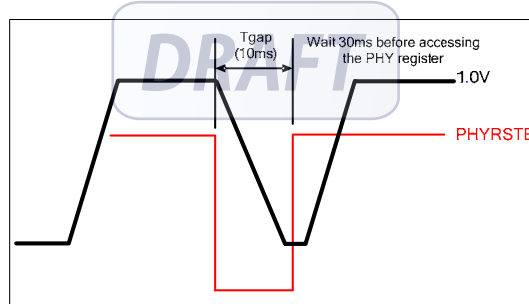


Figure 9. PHY Reset Timing

8. Register Descriptions

8.1. Register Mapping and Definitions

Table 18. Register Access Types

Type	Description
LH	Latch high. If the status is high, this field is set to '1' and remains set.
RC	Read-cleared. The register field is cleared after read.
RO	Read only.
RW	Read and Write
SC	Self-cleared. Writing a '1' to this register field causes the function to be activated immediately, and then the field will be automatically cleared to '0'.

Table 19. Register Mapping and Definitions

Offset	Access	Name	Description
0	RW	BMCR	Basic Mode Control Register.
1	RO	BMSR	Basic Mode Status Register.
2	RO	PHYID1	PHY Identifier Register 1.
3	RO	PHYID2	PHY Identifier Register 2.
4	RW	ANAR	Auto-Negotiation Advertising Register.
5	RO	ANLPAR	Auto-Negotiation Link Partner Ability Register.
6	RO	ANER	Auto-Negotiation Expansion Register.
7	RW	ANNPTR	Auto-Negotiation Next Page Transmit Register.
8	RO	ANNPRR	Auto-Negotiation Next Page Receive Register.
9	RW	GBCR	1000Base-T Control Register.
10	RO	GBSR	1000Base-T Status Register.
11~12	RO	RSVD	Reserved.
13	WO	MACR	MMD Access Control Register.
14	RW	MAADR	MMD Access Address Data Register.
15	RO	GBESR	1000Base-T Extended Status Register.
16~17	RO	RSVD	Reserved.
18	RW	INER	Interrupt Enable Register.
19~23	RO	RSVD	Reserved.
24	RW	PHYCR1	PHY Specific Control Register 1.
25	RW	PHYCR2	PHY Specific Control Register 2.
26	RO	PHYSR	PHY Specific Status Register.
27~28	RO	RSVD	Reserved.
29	RO	INSR	Interrupt Status Register.
30	RO	RSVD	Reserved.
31	RW	PAGSR	Page Select Register.

8.2. MMD Register Mapping and Definitions

Table 20. MMD Register Mapping and Definitions

Device	Offset	Access	Name	Description
3	0	RW	PC1R	PCS Control 1 Register.
3	1	RW	PS1R	PCS Status 1 Register.
3	20	RO	EEECR	EEE Capability Register.
3	22	RC	EEEWER	EEE Wake Error Register.
7	60	RW	EEEAR	EEE Advertisement Register.
7	61	RO	EEELPAR	EEE Link Partner Ability Register.

8.3. Other Page Register Mapping and Definitions

Table 21. Other Page Register Mapping and Definitions

Page	Offset	Access	Name	Description
d04	16	RW	LCR	LED Control Register.
d04	17	RW	EEELCR	EEE LED Control Register.

Note: To switch to Page 0xd04, set Register 31 Data=0x0d04 (change to Page 0xd04). After LED setting, switch to the PHY's Page 0 (Register 31 Data=0000).

8.4. Register Tables

8.4.1. BMCR (Basic Mode Control Register, Address 0x00)

Table 22. BMCR (Basic Mode Control Register, Address 0x00)

Bit	Name	Type	Default	Description
0.15	Reset	RW, SC	0	Reset. 1: PHY reset 0: Normal operation Register 0 (BMCR) and register 1 (BMSR) will return to default values after a software reset (set Bit15 to 1). This action may change the internal PHY state and the state of the physical link associated with the PHY.
0.14	Loopback	RW	0	Loopback Mode. 1: Enable PCS loopback mode 0: Disable PCS loopback mode The loopback function enables RGMII transmit data to be routed to the RGMII receive data path.

Bit	Name	Type	Default	Description															
0.13	Speed[0]	RW	0	Speed Select Bit 0. In forced mode, i.e., when Auto-Negotiation is disabled, bits 6 and 13 determine device speed selection. <table><tr><th>Speed[1]</th><th>Speed[0]</th><th>Speed Enabled</th></tr><tr><td>1</td><td>1</td><td>Reserved</td></tr><tr><td>1</td><td>0</td><td>1000Mbps</td></tr><tr><td>0</td><td>1</td><td>100Mbps</td></tr><tr><td>0</td><td>0</td><td>10Mbps</td></tr></table>	Speed[1]	Speed[0]	Speed Enabled	1	1	Reserved	1	0	1000Mbps	0	1	100Mbps	0	0	10Mbps
Speed[1]	Speed[0]	Speed Enabled																	
1	1	Reserved																	
1	0	1000Mbps																	
0	1	100Mbps																	
0	0	10Mbps																	
0.12	ANE	RW	1	Auto-Negotiation Enable. 1: Enable Auto-Negotiation 0: Disable Auto-Negotiation															
0.11	PWD	RW	0	Power Down. 1: Power down (only Management Interface and logic are active; link is down) 0: Normal operation															
0.10	Isolate	RW	0	Isolate. 1: RGMII interface is isolated; the serial management interface (MDC, MDIO) is still active. When this bit is asserted, the RTL8211F(I)/RTL8211FD(I) ignores TXD[3:0], and TXCTL inputs, and presents a high impedance on TXC, RXC, RXCTL, RXD[3:0]. 0: Normal operation															
0.9	Restart_AN	RW, SC	0	Restart Auto-Negotiation. 1: Restart Auto-Negotiation 0: Normal operation															
0.8	Duplex	RW	1	Duplex Mode. 1: Full Duplex operation 0: Half Duplex operation This bit is valid only in force mode, i.e., NWay is disabled.															
0.7	Collision Test	RW	0	Collision Test. 1: Collision test enabled 0: Normal operation															
0.6	Speed[1]	RW	1	Speed Select Bit 1. Refer to bit 0.13.															
0.5	Uni-directional enable	RW	0	Uni-Directional Enable 1: Enable packet transmit without respect to link status 0: Packet transmit permitted when link is established															
0.4:0	RSVD	RO	00000	Reserved.															

Note 1: When the RTL8211F(I)/RTL8211FD(I) is switched from power down to normal operation, a software reset and restart auto-negotiation is performed, even if bits Reset (0.15) and Restart_AN (0.9) are not set by the user.

Note 2: Auto-Negotiation is enabled when speed is set to 1000Base-T. Crossover Detection & Auto-Correction takes precedence over Auto-Negotiation disable (0.12=0). If ANE is disabled, speed and duplex capabilities are advertised by 0.13, 0.6, and 0.8. Otherwise, register 4.8:5 and 9.9:8 take effect.

Note 3: Auto-Negotiation automatically restarts after hardware or software reset regardless of whether or not the restart bit (0.9) is set.

8.4.2. BMSR (Basic Mode Status Register, Address 0x01)

Table 23. BMSR (Basic Mode Status Register, Address 0x01)

Bit	Name	Type	Default	Description
1.15	100Base-T4	RO	0	100Base-T4 Capability. The RTL8211F(I)/RTL8211FD(I) does not support 100Base-T4 mode. This bit should always be 0.
1.14	100Base-TX (full)	RO	1	100Base-TX Full Duplex Capability. 1: Device is able to perform 100Base-TX in full duplex mode 0: Device is not able to perform 100Base-TX in full duplex mode
1.13	100Base-TX (half)	RO	1	100Base-TX Half Duplex Capability. 1: Device is able to perform 100Base-TX in half duplex mode 0: Device is not able to perform 100Base-TX in half duplex mode
1.12	10Base-T (full)	RO	1	10Base-T Full Duplex Capability. 1: Device is able to perform 10Base-T in full duplex mode. 0: Device is not able to perform 10Base-T in full duplex mode.
1.11	10Base-T (half)	RO	1	10Base-T Half Duplex Capability. 1: Device is able to perform 10Base-T in half duplex mode 0: Device is not able to perform 10Base-T in half duplex mode
1.10	10Base-T2 (full)	RO	0	10Base-T2 Full Duplex Capability. The RTL8211F(I)/RTL8211FD(I) does not support 10Base-T2 mode and this bit should always be 0.
1.9	10Base-T2 (half)	RO	0	10Base-T2 Half Duplex Capability. The RTL8211F(I)/RTL8211FD(I) does not support 10Base-T2 mode. This bit should always be 0.
1.8	1000Base-T Extended Status	RO	1	1000Base-T Extended Status Register. 1: Device supports Extended Status Register 0x0F (15) 0: Device does not support Extended Status Register 0x0F This register is read-only and is always set to 1.
1.7	Uni-directional ability	RO	1	Uni-directional ability. 1: PHY able to transmit from RGMII without linkok 0: PHY not able to transmit from RGMII without linkok
1.6	Preamble Suppression	RO	1	Preamble Suppression Capability (Permanently On). The RTL8211F(I)/RTL8211FD(I) always accepts transactions with preamble suppressed.
1.5	Auto-Negotiation Complete	RO	0	Auto-Negotiation Complete. 1: Auto-Negotiation process complete, and contents of registers 5, 6, 8, and 10 are valid 0: Auto-Negotiation process not complete
1.4	Remote Fault	RC, LH	0	Remote Fault. 1: Remote fault condition detected (cleared on read or by reset). Indication or notification of remote fault from Link Partner 0: No remote fault condition detected
1.3	Auto-Negotiation Ability	RO	1	Auto Configured Link. 1: Device is able to perform Auto-Negotiation 0: Device is not able to perform Auto-Negotiation

Bit	Name	Type	Default	Description
1.2	Link Status	RO	0	Link Status. 1: Linked 0: Not Linked This register indicates whether the link was lost since the last read. For the current link status, either read this register twice or read register bit 17,10 Link Real Time.
1.1	Jabber Detect	RC, LH	0	Jabber Detect. 1: Jabber condition detected 0: No Jabber occurred
1.0	Extended Capability	RO	1	1: Extended register capabilities, always 1

8.4.3. PHYID1 (PHY Identifier Register 1, Address 0x02)

Table 24. PHYID1 (PHY Identifier Register 1, Address 0x02)

Bit	Name	Type	Default	Description
2.15:0	OUI_MSB	RO	0000000000011100	Organizationally Unique Identifier Bit 3:18. Always 0000000000011100.

Note: Realtek OUI is 0x000732.

8.4.4. PHYID2 (PHY Identifier Register 2, Address 0x03)

Table 25. PHYID2 (PHY Identifier Register 2, Address 0x03)

Bit	Name	Type	Default	Description
3.15:10	OUI_LSB	RO	110010	Organizationally Unique Identifier Bit 19:24. Always 110010.
3.9:4	Model Number	RO	010001	Manufacture's Model Number
3.3:0	Revision Number	RO	0110	Revision Number

8.4.5. ANAR (Auto-Negotiation Advertising Register, Address 0x04)

Table 26. ANAR (Auto-Negotiation Advertising Register, Address 0x04)

Bit	Name	Type	Default	Description
4.15	NextPage	RW	0	1: Additional next pages exchange desired 0: No additional next pages exchange desired
4.14	RSVD	RO	0	Reserved.
4.13	Remote Fault	RW	0	1: Set Remote Fault bit 0: No remote fault detected
4.12	RSVD	RO	0	Reserved.
4.11	Asymmetric PAUSE	RW	0	1: Advertise support of asymmetric pause 0: No support of asymmetric pause

Bit	Name	Type	Default	Description
4.10	PAUSE	RW	0	1: Advertise support of pause frames 0: No support of pause frames
4.9	100Base-T4	RO	0	1: 100Base-T4 support 0: 100Base-T4 not supported
4.8	100Base-TX (Full)	RW	1	1: Advertise support of 100Base-TX full-duplex mode 0: Not advertised
4.7	100Base-TX (Half)	RW	1	1: Advertise support of 100Base-TX half-duplex mode 0: Not advertised
4.6	10Base-T (Full)	RW	1	1: Advertise support of 10Base-TX full-duplex mode 0: Not advertised
4.5	10Base-T (Half)	RW	1	1: Advertise support of 10Base-TX half-duplex mode 0: Not advertised
4.4:0	Selector Field	RO	00001	Indicates the RTL8211F(I)/RTL8211FD(I) supports IEEE 802.3

Note 1: The setting of Register 4 has no effect unless NWay is restarted or the link goes down, i.e., software reset (0.15) is asserted, Restart AN (0.9) is asserted, or PWD (0.11) transitions from power down to normal operation.

Note 2: If 1000Base-T is advertised, then the required next pages are automatically transmitted. Register 4.15 should be set to 0 if no additional next pages are needed.

8.4.6. ANLPAR (Auto-Negotiation Link Partner Ability Register, Address 0x05)

Table 27. ANLPAR (Auto-Negotiation Link Partner Ability Register, Address 0x05)

Bit	Name	Type	Default	Description
5.15	Next Page	RO	0	Next Page Indication. Received Code Word Bit 15.
5.14	ACK	RO	0	Acknowledge. Received Code Word Bit 14.
5.13	Remote Fault	RO	0	Remote Fault indicated by Link Partner. Received Code Word Bit 13.
5.12	RSVD	RO	0	Reserved.
5.11:5	Technology Ability Field	RO	00000000	Received Code Word Bit 12:5.
5.4:0	Selector Field	RO	00000	Received Code Word Bit 4:0.

Note: Register 5 is not valid until the Auto-Negotiation complete bit 1.5 indicates completed.

8.4.7. ANER (Auto-Negotiation Expansion Register, Address 0x06)

Table 28. ANER (Auto-Negotiation Expansion Register, Address 0x06)

Bit	Name	Type	Default	Description
6.15:5	RSVD	RO	0x000	Reserved.
6.6	RX NP location ability	RO	1	Received next page storage location ability. 1: Received next page storage location is specified by bit 6.5 0: Received next page storage location is not specified by bit 6.5
6.5	RX NP location	RO	1	Received next page storage location. 1: Link partner next pages are stored in Register 8 0: Link partner next pages are stored in Register 5
6.4	Parallel Detection Fault	RC, LH	0	1: A fault has been detected via the Parallel Detection function 0: A fault has not been detected via the Parallel Detection function
6.3	Link Partner Next Page Able	RO	0	1: Link Partner supports Next Page exchange 0: Link Partner does not support Next Page exchange
6.2	Local Next Page Able	RO	1	1: Local Device is able to send Next Page Always 1.
6.1	Page Received	RC, LH	0	1: A New Page (new LCW) has been received 0: A New Page has not been received
6.0	Link Partner Auto-Negotiation capable	RO	0	1: Link Partner supports Auto-Negotiation 0: Link Partner does not support Auto-Negotiation

Note: Register 6 is not valid until the Auto-Negotiation complete bit 1.5 indicates completed.

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8.4.8. ANNPTR (Auto-Negotiation Next Page Transmit Register, Address 0x07)

Table 29. ANNPTR (Auto-Negotiation Next Page Transmit Register, Address 0x07)

Bit	Name	Type	Default	Description
7.15	Next Page	RW	0	Next Page Indication. 0: No more next pages to send 1: More next pages to send Transmit Code Word Bit 15.
7.14	RSVD	RO	0	Transmit Code Word Bit 14.
7.13	Message Page	RW	1	Message Page. 0: Unformatted Page 1: Message Page Transmit Code Word Bit 13.
7.12	Acknowledge 2	RW	0	Acknowledge2. 0: Local device has no ability to comply with the message received 1: Local device has the ability to comply with the message received Transmit Code Word Bit 12.
7.11	Toggle	RO	0	Toggle Bit. Transmit Code Word Bit 11.
7.10:0	Message/Unformatted Field	RW	0x001	Content of Message/Unformatted Page. Transmit Code Word Bit 10:0.

8.4.9. ANNPRR (Auto-Negotiation Next Page Receive Register, Address 0x08)

Table 30. ANNPRR (Auto-Negotiation Next Page Receive Register, Address 0x08)

Bit	Name	Type	Default	Description
8.15	Next Page	RO	0	Received Link Code Word Bit 15.
8.14	Acknowledge	RO	0	Received Link Code Word Bit 14.
8.13	Message Page	RO	0	Received Link Code Word Bit 13.
8.12	Acknowledge 2	RO	0	Received Link Code Word Bit 12.
8.11	Toggle	RO	0	Received Link Code Word Bit 11.
8.10:0	Message/Unformatted Field	RO	0x00	Received Link Code Word Bit 10:0.

Note: Register 8 is not valid until the Auto-Negotiation complete bit 1.5 indicates completed.

8.4.10. GBCR (1000Base-T Control Register, Address 0x09)

Table 31. GBCR (1000Base-T Control Register, Address 0x09)

Bit	Name	Type	Default	Description
9.15:13	Test Mode	RW	0	Test Mode Select. 000: Normal Mode 001: Test Mode 1 – Transmit Jitter Test 010: Test Mode 2 – Transmit Jitter Test (MASTER mode) 011: Test Mode 3 – Transmit Jitter Test (SLAVE mode) 100: Test Mode 4 – Transmit Distortion Test 101, 110, 111: Reserved
9.12	MASTER/SLAVE Manual Configuration Enable	RW	0	Enable Manual Master/Slave Configuration. 1: Manual MASTER/SLAVE configuration 0: Automatic MASTER/SLAVE
9.11	MASTER/SLAVE Configuration Value	RW	0	Advertise Master/Slave Configuration Value. 1: Manual configure as MASTER 0: Manual configure as SLAVE
9.10	Port Type	RW	0	Advertise Device Type Preference. 1: Prefer multi-port device (MASTER) 0: Prefer single port device (SLAVE)
9.9	1000Base-T Full Duplex	RW	1	Advertise 1000Base-T Full-Duplex Capability. 1: Advertise 0: Do not advertise
9.8	RSVD	RW	0	Reserved.
9.7:0	RSVD	RO	0	Reserved.

Note 1: Values set in register 9.12:9 have no effect unless Auto-Negotiation is restarted (Reg0.9) or the link goes down.

Note 2: Bits 9.11 and 9.10 are ignored when bit 9.12=0.

8.4.11. GBSR (1000Base-T Status Register, Address 0x0A)

Table 32. GBSR (1000Base-T Status Register, Address 0x0A)

Bit	Name	Type	Default	Description
10.15	MASTER/SLAVE Configuration Fault	RO, RC, LH	0	Master/Slave Manual Configuration Fault Detected. 1: MASTER/SLAVE configuration fault detected 0: No MASTER/SLAVE configuration fault detected
10.14	MASTER/SLAVE Configuration Resolution	RO	0	Master/Slave Configuration Result. 1: Local PHY configuration resolved to MASTER 0: Local PHY configuration resolved to SLAVE
10.13	Local Receiver Status	RO	0	Local Receiver Status. 1: Local Receiver OK 0: Local Receiver Not OK
10.12	Remote Receiver Status	RO	0	Remote Receiver Status. 1: Remote Receiver OK 0: Remote Receiver Not OK

Bit	Name	Type	Default	Description
10.11	Link Partner 1000Base-T Full Duplex Capability	RO	0	Link Partner 1000Base-T Full Duplex Capability. 1: Link Partner is capable of 1000Base-T full duplex 0: Link Partner is not capable of 1000Base-T full duplex
10.10	Link Partner 1000Base-T Half Duplex Capability	RO	0	Link Partner 1000Base-T Half Duplex Capability. 1: Link Partner is capable of 1000Base-T half duplex 0: Link Partner is not capable of 1000Base-T half duplex
10.9:8	RSVD	RO	00	Reserved.
10.7:0	Idle Error Count	RO, RC	0x00	MSB of Idle Error Counter. The counter stops automatically when it reaches 0xff.

Note 1: Values set in register 10.11:10 are not valid until register 6.1 is set to 1.

Note 2: Register 10 is not valid until the Auto-Negotiation complete bit 1.5 indicates completed.

8.4.12. MACR (MMD Access Control Register, Address 0x0D)

Table 33. MACR (MMD Access Control Register, Address 0x0D)

Bit	Name	Type	Default	Description
13.15:14	Function	WO	0	00: Address 01: Data with no post increment 10: Data with post increment on reads and writes 11: Data with post increment on writes only
13.13:5	RSVD	RO	000000000	Reserved.
13.4:0	DEVAD	WO	0	Device Address.

Note 1: This register is used in conjunction with the MAADR (Register 14) to provide access to the MMD address space.

Note 2: If the MAADR accesses for address (Function=00), then it is directed to the address register within the MMD associated with the value in the DEVAD field.

Note 3: If the MAADR accesses for data (Function≠00), both the DEVAD field and MMD's address register direct the MAADR data accesses to the appropriate registers within the MMD.

8.4.13. MAADR (MMD Access Address Data Register, Address 0x0E)

Table 34. MAADR (MMD Access Address Data Register, Address 0x0E)

Bit	Name	Type	Default	Description
14.15:0	Address Data	RW	0x0000	13.15:14 = 00 → MMD DEVAD's address register 13.15:14 = 01, 10, or 11 → MMD DEVAD's data register as indicated by the contents of its address register

Note: This register is used in conjunction with the MACR (Register 13; Table 33) to provide access to the MMD address space.

8.4.14. GBESR (1000Base-T Extended Status Register, Address 0x0F)

Table 35. GBESR (1000Base-T Extended Status Register, Address 0x0F)

Bit	Name	Type	Default	Description
15.15	1000Base-X FD	RO	0	0: Not 1000Base-X full duplex capable
15.14	1000Base-X HD	RO	0	0: Not 1000Base-X half duplex capable
15.13	1000Base-T FD	RO	1	1: 1000Base-T full duplex capable
15.12	1000Base-T HD	RO	0	1: 1000Base-T half duplex capable
15.11:0	RSVD	RO	0x000	Reserved.

8.4.15. INER (Interrupt Enable Register, Address 0x12)

Table 36. INER (Interrupt Enable Register, Address 0x12)

Bit	Name	Type	Default	Description
18.15:12	RSVD	RW	00000	Reserved.
18.10	Jabber Interrupt	RW	0	1: Interrupt Enable 0: Interrupt Disable Setting this bit to 0 only masks a jabber interrupt event in the INT interface. Reg29 Bit10 always reflects the jabber interrupt behavior.
18.9	ALDPS State Change Interrupt	RW	0	1: Interrupt Enable 0: Interrupt Disable Setting this bit to 0 only masks an ALDPS state change interrupt event in the INT interface. Reg29 Bit9 always reflects the ALDPS state change interrupt behavior.
18.8	RSVD	RW	0	Reserved.
18.7	PME (Power Management Event of WOL)	RW	0	1: Interrupt Enable 0: Interrupt Disable Setting this bit to 0 only masks a PME interrupt event in the INT interface. Reg29 Bit7 always reflects the PME interrupt behavior.
18.6	RSVD	RW	0	Reserved.
18.5	PHY Register Access Interrupt	RW	0	1: Interrupt Enable 0: Interrupt Disable Setting this bit to 0 only masks a PHY register access interrupt event in the INT interface. Reg29 Bit12 always reflects the PHY register access interrupt behavior.
18.4	Link Status Change Interrupt	RW	0	1: Interrupt Enable 0: Interrupt Disable Setting this bit to 0 only masks a link status change interrupt event in the INT interface. Reg29 Bit4 always reflects the link change interrupt behavior.
18.3	Auto-Negotiation Completed Interrupt	RW	0	1: Interrupt Enable 0: Interrupt Disable Setting this bit to 0 only masks an auto-negotiation completed interrupt event in the INT interface. Reg29 Bit3 always reflects the auto-negotiation completed interrupt behavior.

Bit	Name	Type	Default	Description
18.2	Page Received Interrupt	RW	0	1: Interrupt Enable 0: Interrupt Disable Setting this bit to 0 only masks a page received interrupt event in the INT interface. Reg29 Bit2 always reflects the page received interrupt behavior.
18.1	RSVD	RW	0	Reserved.
18.0	Auto-Negotiation Error Interrupt	RW	0	1: Interrupt Enable 0: Interrupt Disable Setting this bit to 0 only masks an auto-negotiation error interrupt event in the INT interface. Reg29 Bit0 always reflects the auto-negotiation error interrupt behavior.

8.4.16. PHYCR1 (PHY Specific Control Register 1, Address 0x18)

Table 37. PHYCR1 (PHY Specific Control Register 1, Address 0x18)

Bit	Name	Type	Default	Description
24.15:14	RSVD	RO	00	Reserved.
24.13	PHYAD_0 Enable	RW	1	1: A broadcast from MAC (A command with PHY address = 0) is valid. MDC/MDIO will respond to this command.
24.12:10	RSVD	RO	000	Reserved.
24.9	MDI Mode Manual Configuration Enable	RW	0	1: Enable Manual Configuration of MDI mode
24.8	MDI Mode	RW	1	Set the MDI/MDIX mode. 1: MDI 0: MDIX This bit will take effect only when Reg 24.9 = 1.
24.7	TX CRS Enable	RW	0	1: Assert CRS on transmit 0: Never assert CRS on transmit
24.6	PHYAD Non-zero Detect	RW	0	1: The RTL8211F(I)/RTL8211FD(I) with PHYAD[2:0] = 000 will latch the first non-zero PHY address as its own PHY address
24.5	RSVD	RO	0	Reserved.
24.4	Preamble Check Enable	RW	1	1: Check preamble when receiving an MDC/MDIO command
24.3	Jabber Detection Enable	RW	1	1: Enable Jabber Detection
24.2	ALDPS Enable	RO	0	1: Enable Link Down Power Saving Mode
24.1:0	RSVD	RO	00	Reserved.

Note: The method to disable auto-crossover and force MDI or MDIX mode is as follows:

Step 1: Set Reg24 bit[9]=1 (Manual Configuration of MDI mode) and set Reg24 bit[8]=1 (MDI) or 0 (MDIX).

Step 2: Perform a PHY reset, i.e., set Reg0 bit[15]=1.

8.4.17. PHYCR2 (PHY Specific Control Register 2, Address 0x19)

Table 38. PHYCR2 (PHY Specific Control Register 2, Address 0x19)

Bit	Name	Type	Default	Description
25.15:12	RSVD	RO	0000	Reserved.
25.11	CLKOUT Frequency Select	RW	1	Frequency select of the CLKOUT pin clock output. 0: 25MHz 1: 125MHz
25.10:8	RSVD	RO	000	Reserved.
25.7	CLKOUT SSC Enable	RW	0	1: Enable Spread-Spectrum Clocking (SSC) on CLKOUT output clock.
25.6:4	RSVD	RO	000	Reserved.
25.3	RXC SSC Enable	RW	0	1: Enable Spread-Spectrum Clocking (SSC) on RXC clock output.
25.2	RSVD	RO	0	Reserved.
25.1	RXC Enable	RO	0	1: RXC clock output enabled.
25.0	CLKOUT Enable	RW	1	1: CLKOUT clock output enabled.

8.4.18. PHYSR (PHY Specific Status Register, Address 0x1A)

Table 39. PHYSR (PHY Specific Status Register, Address 0x1A)

Bit	Name	Type	Default	Description
26.15	RSVD	RO	0	Reserved.
26.14	ALDPS State	RO	0	Link Down Power Saving Mode. 1: Reflects local device entered Link Down Power Saving Mode, i.e., cable not plugged in (reflected after 3 sec). 0: With cable plugged in
26.13	MDI Plug	RO	0	MDI Status. 1: Plugged 0: Unplugged
26.12	NWay Enable	RO	1	Auto-Negotiation (NWay) Status. 1: Enable 0: Disable
26.11	Master Mode	RO	0	Device is in Master/Slave Mode. 1: Master mode 0: Slave mode
26.10:9	RSVD	RO	00	Reserved.
26.8	EEE capability	RO	0	1: Both local and link-partner have EEE capability of current speed
26.7	Rxflow Enable	RO	0	Rx Flow Control. 1: Enable 0: Disable
26.6	Txflow Enable	RO	0	Tx Flow Control. 1: Enable 0: Disable
26.5:4	Speed	RO	00	Link Speed. 11: Reserved 10: 1000Mbps 01: 100Mbps 00: 10Mbps
26.3	Duplex	RO	0	Full/Half Duplex Mode. 1: Full duplex 0: Half duplex
26.2	Link (Real Time)	RO	0	Real Time Link Status. 1: Link OK 0: Link not OK

Bit	Name	Type	Default	Description
26.1	MDI Crossover Status	RO	1	MDI/MDI Crossover Status. 1: MDI 0: MDI Crossover
26.0	Jabber (Real Time)	RO	0	Real Time Jabber Indication. 1: Jabber Indication 0: No Jabber Indication

Note 1: Bit 26.11 valid only when in Giga mode.

Note 2: Bit 26.8 assert at 10M speed when local device is EEE capable.

8.4.19. INSR (Interrupt Status Register, Address 0x13)

Table 40. INSR (Interrupt Status Register, Address 0x13)

Bit	Name	Type	Default	Description
29.15:11	RSVD	RO, RC	00000	Reserved.
29.10	Jabber	RO, RC	0	1: Jabber detected 0: No jabber detected
29.9	ALDPS State Change	RO, RC	0	1: ALDPS state changed 0: ALDPS state not changed
29.8	RSVD	RO, RC	0	Reserved.
29.7	PME (Power Management Event of WOL)	RO, RC	0	1: WOL event occurred 0: WOL event did not occur
29.6	RSVD	RO, RC	0	Reserved.
29.5	PHY Register Access	RO, RC	0	1: Can access PHY Register through MDC/MDIO 0: Cannot access PHY Register through MDC/MDIO
29.4	Link Status Change	RO, RC	0	1: Link status changed 0: Link status not changed
29.3	Auto-Negotiation Completed	RO, RC	0	1: Auto-Negotiation completed 0: Auto-Negotiation not completed
29.2	Page Received	RO, RC	0	1: Page (a new LCW) received 0: Page not received
29.1	RSVD	RO, RC	0	Reserved.
29.0	Auto-Negotiation Error	RO, RC	0	1: Auto-Negotiation Error 0: No Auto-Negotiation Error

8.4.20. EPAGSR (Extension Page Select Register, Address 0x1F)

Table 41. EPAGSR (Extension Page Select Register, Address 0x1F)

Bit	Name	Type	Default	Description
31.15:12	RSVD	RW	0	Reserved.
31.11:0	ExtPageSel	RW	0x000	Page Select (in HEX). 0x000: Page 0 (default page)

8.4.21. PC1R (PCS Control 1 Register, MMD Device 3, Address 0x00)

Table 42. PC1R (PCS Control 1 Register, MMD Device 3, Address 0x00)

Bit	Name	Type	Default	Description
3.0.15:11	RSVD	RW	0	Reserved.
3.0.10	Clock Stop Enable	RW	0	1: PHY stops RXC in LPI 0: RXC not stoppable
3.0.9:0	RSVD	RW	0	Reserved.

8.4.22. PS1R (PCS Status1 Register, MMD Device 3, Address 0x01)

Table 43. PS1R (PCS Status 1 Register, MMD Device 3, Address 0x01)

Bit	Name	Type	Default	Description
3.1.15:12	RSVD	RO	0	Reserved.
3.1.11	TX LPI Received	RO, LH	0	1: TX PCS has received LPI 0: LPI not received
3.1.10	RX LPI Received	RO, LH	0	1: RX PCS has received LPI 0: LPI not received
3.1.9	TX LPI Indication	RO	0	1: TX PCS is currently receiving LPI 0: TX PCS is not currently receiving LPI
3.1.8	RX LPI Indication	RO	0	1: RX PCS is currently receiving LPI 0: RX PCS is not currently receiving LPI
3.1.7	RSVD	RO	0	Reserved.
3.1.6	Clock Stop Capable	RO	1	1: MAC stops TXC in LPI 0: TXC not stoppable
3.1.5:0	RSVD	RO	0	Reserved.

8.4.23. EEECR (EEE Capability Register, MMD Device 3, Address 0x14)

Table 44. EEECR (EEE Capability Register, MMD Device 3, Address 0x14)

Bit	Name	Type	Default	Description
3.20.15:3	RSVD	RO	0	Reserved.
3.20.2	1000BASE-T EEE	RO	1	1: EEE is supported for 1000Base-T EEE 0: EEE is not supported for 1000Base-T EEE
3.20.1	100BASE-TX EEE	RO	1	1: EEE is supported for 100Base-TX EEE 0: EEE is not supported for 100Base-TX EEE
3.20.0	RSVD	RO	0	Reserved.

8.4.24. EEWER (EEE Wake Error Register, MMD Device 3, Address 0x16)

Table 45. EEWER (EEE Wake Error Register, MMD Device 3, Address 0x16)

Bit	Name	Type	Default	Description
3.22.15:0	EEE Wake Error Counter	RC	0	Used by PHY types that support EEE to count wake time faults where the PHY fails to complete its normal wake sequence within the time required for the specific PHY type.

8.4.25. EEER (EEE Advertisement Register, MMD Device 7, Address 0x3c)

Table 46. EEER (EEE Advertisement Register, MMD Device 7, Address 0x3c)

Bit	Name	Type	Default	Description
7.60.15:3	RSVD	RW	0	Reserved.
7.60.2	1000BASE-T EEE	RW	1	Advertise 1000Base-T EEE Capability. 1: Advertise 0: Do not advertise
7.60.1	100BASE-TX EEE	RW	1	Advertise 100Base-TX EEE Capability. 1: Advertise 0: Do not advertise
7.60.0	RSVD	RW	0	Reserved.

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8.4.26. EEELPAR (EEE Link Partner Ability Register, MMD Device 7, Address 0x3d)

Table 47. EEELPAR (EEE Link Partner Ability Register, MMD Device 7, Address 0x3d)

Bit	Name	Type	Default	Description
7.61.15:3	RSVD	RO	0	Reserved.
7.61.2	LP 1000BASE-T EEE	RO	0	1: Link Partner is capable of 1000Base-T EEE 0: Link Partner is not capable of 1000Base-T EEE
7.61.1	LP 100BASE-TX EEE	RO	0	1: Link Partner is capable of 100Base-TX EEE 0: Link Partner is not capable of 100Base-TX EEE
7.61.0	RSVD	RO	0	Reserved.

8.4.27. LCR (LED Control Register, ExtPage 0xd04, Address 0x10)

Table 48. LCR (LED Control Register, ExtPage 0xd04, Address 0x10)

Bit	Name	Type	Default	Description
16.15	RSVD	RO	0	Reserved.
16.14	LED2_ACT	RW	1	LED2 Active (Transmitting or Receiving) Indication
16.13	LED2_LINK_1000	RW	1	LED2 Link Indication: 1000Mbps
16.12	RSVD	RO	0	Reserved.
16.11	LED2_LINK_100	RW	0	LED2 Link Indication: 100Mbps
16.10	LED2_LINK_10	RW	0	LED2 Link Indication: 10Mbps
16.9	LED1_ACT	RW	1	LED1 Active (Transmitting or Receiving) Indication
16.8	LED1_LINK_1000	RW	0	LED1 Link Indication: 1000Mbps
16.7	RSVD	RO	0	Reserved.
16.6	LED1_LINK_100	RW	1	LED1 Link Indication: 100Mbps
16.5	LED1_LINK_10	RW	0	LED1 Link Indication: 10Mbps
16.4	LED0_ACT	RW	1	LED0 Active (Transmitting or Receiving) Indication
16.3	LED0_LINK_1000	RW	0	LED0 Link Indication: 1000Mbps
16.2	RSVD	RO	0	Reserved.
16.1	LED0_LINK_100	RW	0	LED0 Link Indication: 100Mbps
16.0	LED0_LINK_10	RW	1	LED0 Link Indication: 10Mbps

8.4.28. EEELCR (EEE LED Control Register, ExtPage 0xd04, Address 0x11)

Table 49. EEELCR (EEE LED Control Register, ExtPage 0xd04, Address 0x11)

Bit	Name	Type	Default	Description
17.15:4	RSVD	RO	0	Reserved.
17.3	LED2 EEE Enable	RW	1	1: Enable EEE LED indication of LED2.
17.2	LED1 EEE Enable	RW	1	1: Enable EEE LED indication of LED1.
17.1	LED0 EEE Enable	RW	1	1: Enable EEE LED indication of LED0.
17.0	RSVD	RO	0	Reserved.

9. Regulators and Power Sequence

9.1. Switching Regulator (RTL8211F(I))

The RTL8211F(I) incorporates a state-of-the-art switching regulator that requires a well-designed PCB layout in order to achieve good power efficiency and lower the output voltage ripple and input overshoot. The switching regulator 1.0V output pin (REG_OUT) should be connected only to DVDD10 and AVDD10 (do not provide this power source to other devices).

9.1.1. PCB Layout

- The input 3.3V power trace connected to DVDD33 must be wider than 40mils
- The bulk de-coupling capacitors (Cin1 and Cin2) must be placed within 200mils (0.5cm) of DVDD33 to prevent input voltage overshoot
- The output power trace out of REG_OUT must be wider than 60mils
- Lx (2.2 μ H/4.7 μ H) must be kept within 200mils (0.5cm) of REG_OUT
- Cout1 and Cout2 must be kept within 200mils (0.5cm) of Lx to ensure stable output power and better power efficiency
- For switching regulator stability, the capacitor Cout1 and Cout2 must be a ceramic (X5R) capacitor. Cin1 and Cin2 are recommended to be ceramic capacitors
- Place Lx and Cin1 on the same layer as the RTL8211F(I). Do not use vias on DVDD33 and REG_OUT traces

Note: Violation of the above rules will damage the IC.

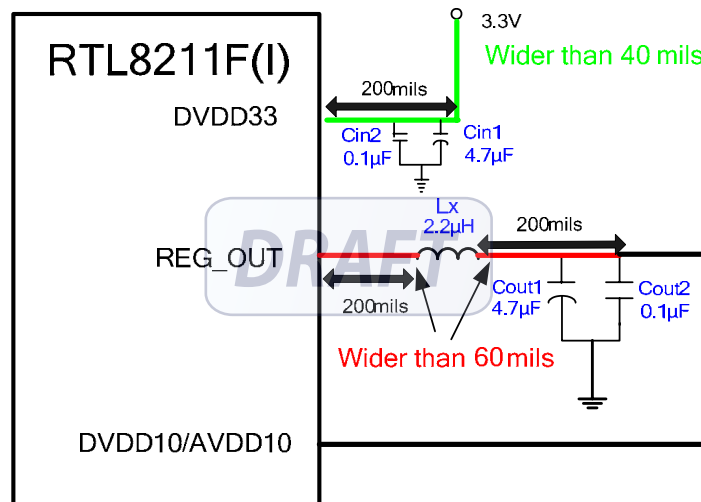


Figure 10. Switching Regulator

9.1.2. Inductor and Capacitor Parts List

Table 50. Inductor and Capacitor Parts List

Inductor Type	Inductance	Max IDC (mA)	Variation	Output Ripple (mV)
GLK2510P-2R2M	2.2 μ H	1000	$\leq 20\%$	(See Figure 14, page 49, and Figure 15, page 50)
GLK2510P-4R7M	4.7 μ H	750	$\leq 20\%$	(See Figure 16, page 50, and Figure 17, page 51)
GTSD32P-2R2M	2.2 μ H	1500	$\leq 20\%$	(See Figure 18, page 51)

Note 1: The ESR is equivalent to RDC or DCR. Lower ESR inductor values will promote a higher-efficiency switching regulator.

Note 2: The power inductor used by the switching regulator must be able to withstand 600mA of current.

Note 3: Typically, if the power inductor's ESR at 1MHz is below 0.8 Ω , the switching regulator efficiency will be above 75%. However the actual switching regulator efficiency should be measured according to the method described in section 9.1.4 Efficiency Measurement, page 54.

Note 4: If the inductor does not meet this requirement, it may damage the switching regulator.

Capacitor Type	Capacitance	ESR at 1MHz (m Ω)	Output Ripple (mV)
4.7 μ F 0805 X5R TDK	4.838	40.28	(See Figure 14, Figure 16)
10 μ F 0603 X5R YAGEO	11.956	58.29	(See Figure 15, Figure 17)

Note: Capacitors (Cin1 & Cin2) must be ceramic due to their low ESR value. Lower ESR values will yield lower output voltage ripple.

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9.1.3. Measurement Criteria

In order for the switching regulator to operate properly, the input and output voltage measurement criteria must be met. From the input side, the voltage overshoot cannot exceed 4V; otherwise the chip may be damaged. Note that the voltage signal must be measured directly at the DVDD33 pin, not at the capacitor. In order to reduce the input voltage overshoot, the Cin1 and Cin2 must be placed close to the DVDD33 pin. The following figures show what a good input voltage and a bad one look like.

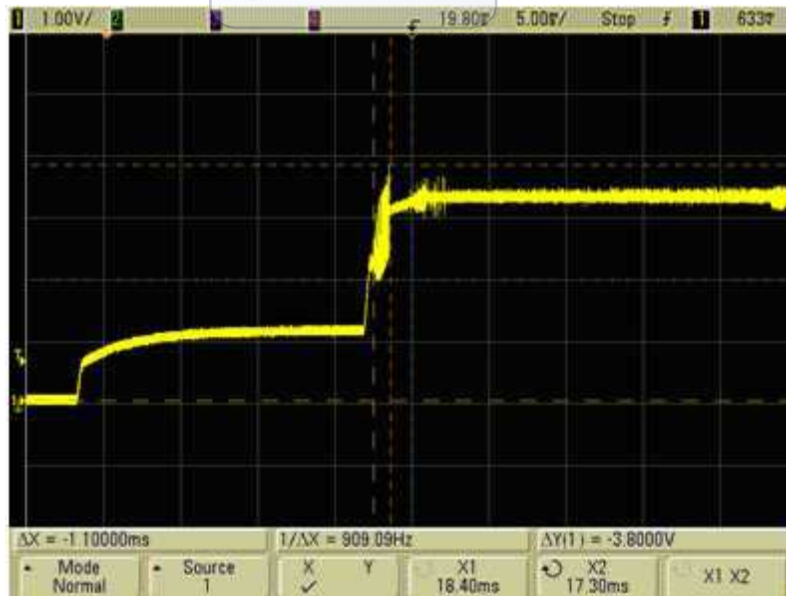


Figure 11. Input Voltage Overshoot <4V (Good)



Figure 12. Input Voltage Overshoot >4V (Bad)

From the output side measured at the REG_OUT pin, the voltage ripple must be within 100mV peak-to-peak. Choosing different types and values of input and output capacitor (Cin1, Cin2; Cout1, Cout2) and power inductor (Lx) will seriously affect the efficiency and output voltage ripple of switching regulators. The following figures show the effects of different types of capacitors on the switching regulator's output voltage.

The blue square wave signal (top row) is measured at the output of the REG_OUT pin before the power inductor (Lx). The yellow signal (second row) is measured after the power inductor (Lx), and shows there is a voltage ripple. The green signal (lower row) is the current. Data in the following figures was measured at gigabit speed.



Figure 13. Ceramic 10 μ F 0603 (X5R) (Good)

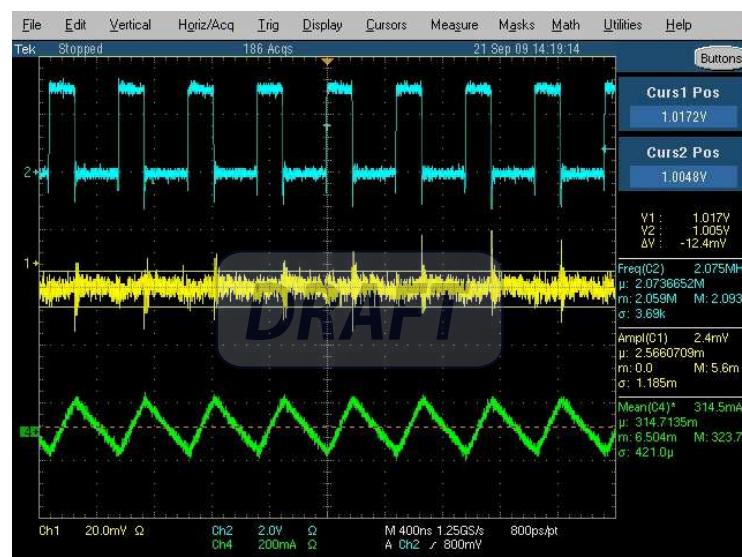


Figure 14. L=GLK2510P-2R2M, C=Ceramic 4.7 μ F 0805 X5R TDK (Ripple 12.4mV)



Figure 15. L=GLK2510P-2R2M, C=Ceramic 10 μ F 0603 X5R YAGEO (Ripple 13.2mV)



Figure 16. L=GLK2510P-4R7M, C=Ceramic 4.7 μ F 0805 X5R TDK (Ripple 12mV)



Figure 17. L=GLK2510P-4R7M, C=Ceramic 10 μ F 0603 X5R YAGEO (Ripple 11.2mV)



Figure 18. L=GTSD32P-2R2M, C=Ceramic 4.7 μ F 0805 X5R TDK (Ripple 9.2mV)

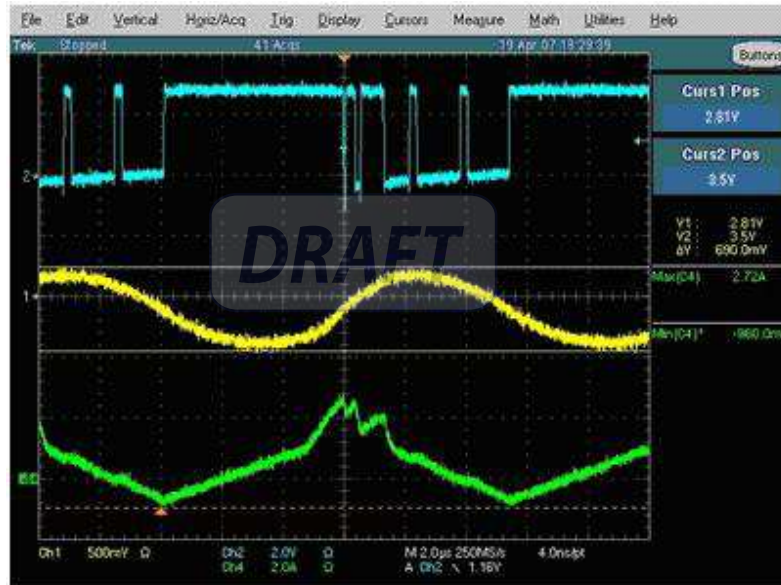


Figure 19. Ceramic 10 μ F (Y5V) (Bad)

A ceramic 10 μ F (X5R) will have a lower voltage ripple compared to an electrolytic 100 μ F. The key to choosing a proper output capacitor is to choose the lowest ESR to reduce the output voltage ripple. Choosing a ceramic 10 μ F (Y5V) in this case will cause malfunction of the switching regulator. Placing several Electrolytic capacitors in parallel will help lower the output voltage ripple.

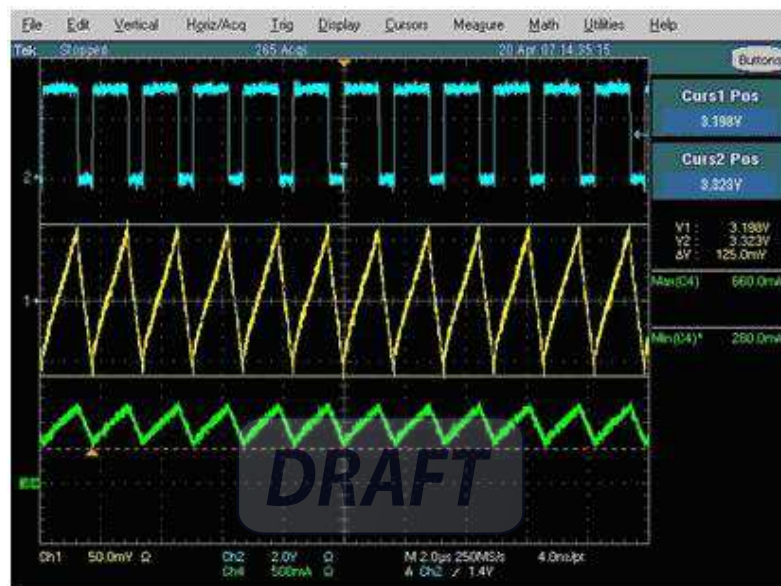


Figure 20. Electrolytic 100 μ F (Ripple Too High)

The following figures show how different inductors affect the REG_OUT output waveform. The typical waveform should look like Figure 21, which has a square waveform with a dip at the falling edge and the rising edge. If the inductor is not carefully chosen, the waveform may look like Figure 22, where the waveform looks like a distorted square. This will cause insufficient current supply and will undermine the stability of the system at gigabit speed. Data in the following figures was measured at gigabit speed

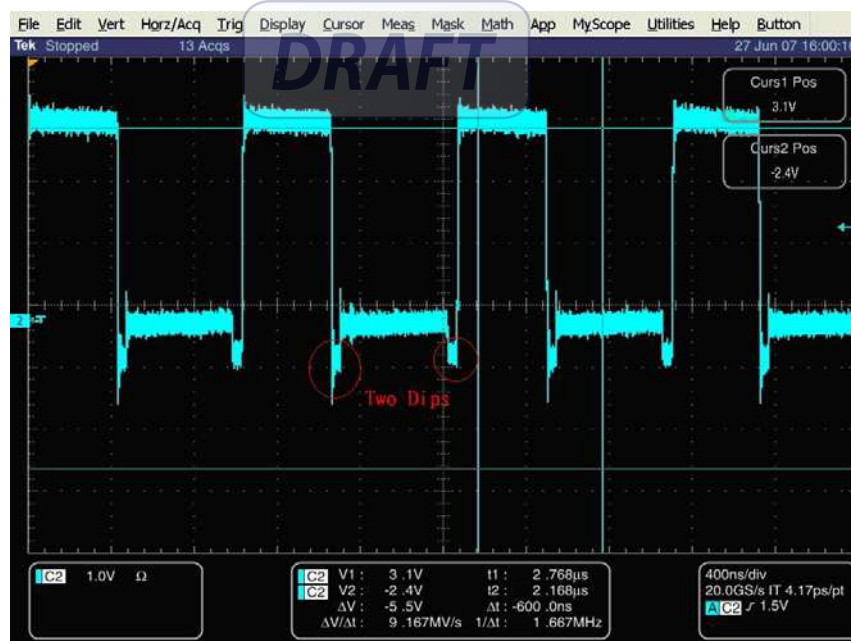


Figure 21. GTSD32P-2R2M (Good)

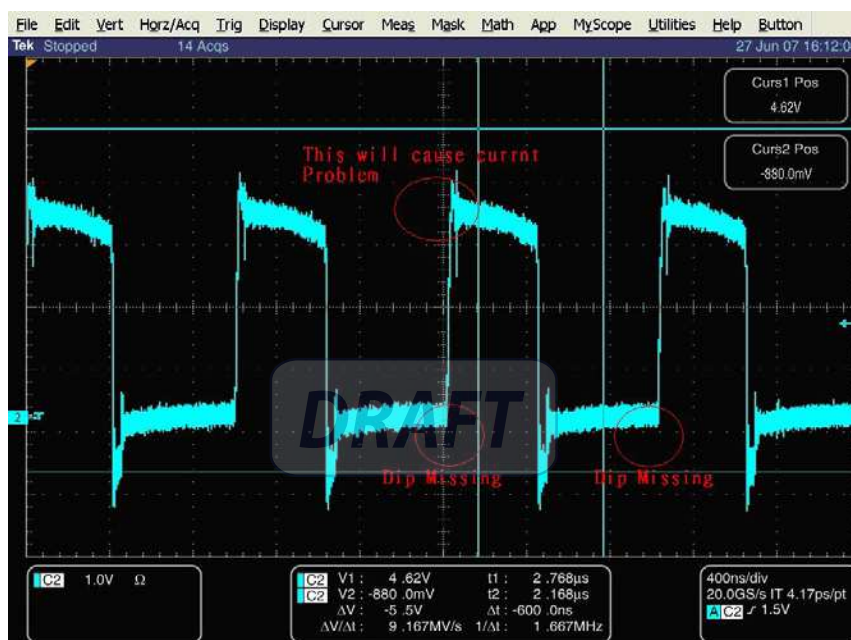


Figure 22. 1μH Bead (Bad)

9.1.4. Efficiency Measurement

The efficiency of the switching regulator is designed to be above 75% in gigabit traffic mode. It is very important to choose a suitable inductor before Gerber certification, as the Inductor ESR value will affect the efficiency of the switching regulator. An inductor with a lower ESR value will result in a higher-efficiency switching regulator.

The efficiency of the switching regulator is easily measured using the following method.

Figure 23 shows two checkpoints, checkpoint A (CP_A) and checkpoint B (CP_B). The switching regulator input current (Icpa) should be measured at CP_A, and the switching regulator output current (Icpb) should be measured at CP_B.

To determine efficiency, apply the following formula:

$$\text{Efficiency} = V_{cpb} \cdot I_{cpb} / V_{cpa} \cdot I_{cpa}$$

Where V_{cpb} is 1.05V; V_{cpa} is 3.3V. The measurements should be performed in gigabit traffic mode.

For example: The inductor used in the evaluation board is a GOTREND GTSD32-4R7M:

- The ESR value @ 1MHz is approximately 0.712ohm
- The measured I_{cpa} is 101mA at CP_A
- The measured I_{cpb} is 263mA at CP_B

These values are measured in gigabit traffic mode, so the efficiency of the GOTREND GTSD32-4R7M can be calculated as follows:

$$\text{Efficiency} = (1.05V \cdot 263mA) / (3.3V \cdot 101mA) = 0.823 = 82.3\%.$$

We strongly recommend that when choosing an inductor for the switching regulator, the efficiency should be measured, and that the inductor should yield an efficiency rating higher than 75%. If the efficiency does not meet this requirement, there may be risk to the switching regulator reliability in the long run.

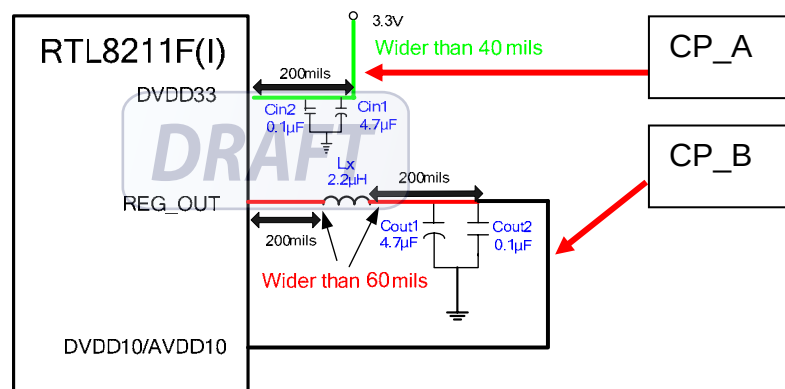


Figure 23. Switching Regulator Efficiency Measurement Checkpoint

9.2. Low-dropout Regulator (RTL8211FD(I))

The RTL8211FD(I) incorporates a state-of-the-art low-dropout regulator (LDO). The regulator 1.0V output pin (LDO_OUT) should be connected only to DVDD10 and AVDD10 pins (do not provide this power source to other devices).

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9.3. Power Sequence

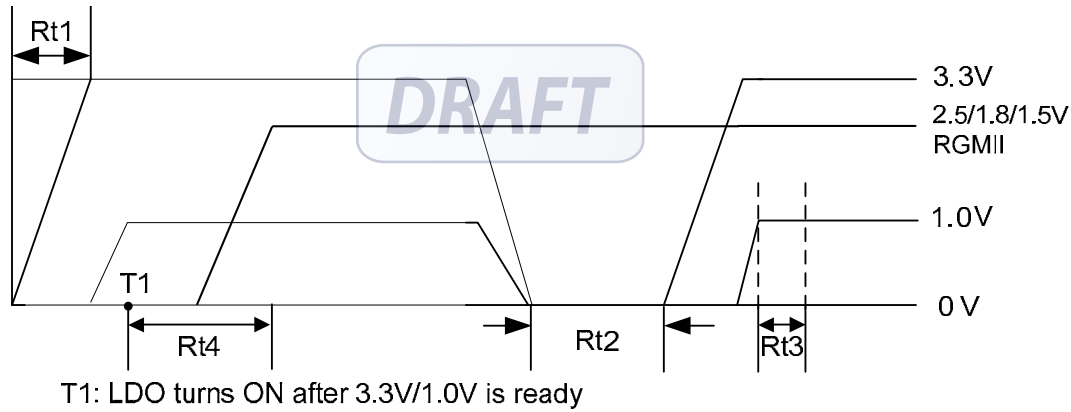


Figure 24. Power Sequence

Table 51. Power Sequence Parameters

Symbol	Description	Min	Typical	Max	Units
Rt1	3.3V Rise Time	0.5*	-	100	ms
Rt1	2.5/1.8/1.5V RGMII Rise Time	-	-	100	ms
Rt2	3.3V Off Time	100	-	-	ms
Rt3	Core Logic Ready Time	20	-	-	ms
Rt4	LDO Ready time	1.5	-	-	ms

Note 1: The RTL8211F(I)/RTL8211FD(I) does not support fast 3.3V rising. The 3.3V rise time should be controlled over 0.5ms.

* A 3.3V rise time between 0.1ms to 0.5ms is conditionally permitted only if the system 3.3V power budget is sufficient to ensure that 3.3V Overcurrent Protection (OCP) will NOT be triggered during the power-on procedure. If the rise time is less than 0.1ms, it will induce a peak voltage in DVDD33 which may cause permanent damage to the regulator.

Note 2: If there is any action that involves consecutive ON/OFF toggling of the switching-regulator source (3.3V), the design must make sure the OFF state of both the switching-regulator source (3.3V) and output (1.05V) reach 0V, and the time period between the consecutive ON/OFF toggling action must be longer than 100ms.

Note 3: When using an external oscillator or clock source, on stopping the clock source the RTL8211F(I)/RTL8211FD(I) must also be powered off.

Note 4: The RTL8211F(I)/RTL8211FD(I) use the integrated LDO to generate the 2.5V, 1.8V/1.5V voltages for the I/O pad, the I/O pad voltage can be selected by using the CONFIG pins CFG_LDO[1:0]. Moreover, external power source for the I/O pad is also supported, please refer to the setting of CFG_EXT and CFG_LDO[1:0] pins (section 7.8 Hardware Configuration, page 15).

Note 5: When using an external power source for the I/O pad, 2.5V (or 1.8/1.5V) RGMII power should rise simultaneously or slightly earlier than 3.3V power. Rising 2.5V (or 1.8/1.5V) power later than 3.3V power may lead to errors.

10. Characteristics

10.1. Absolute Maximum Ratings

WARNING: Absolute maximum ratings are limits beyond which permanent damage may be caused to the device, or device reliability will be affected. All voltages are specified reference to GND unless otherwise specified.

Table 52. Absolute Maximum Ratings

Symbol	Description	Minimum	Maximum	Unit
VDD33, AVDD33	Supply Voltage 3.3V	-0.3	3.6	V
AVDD10, DVDD10	Supply Voltage 1.0V	-0.3	1.2	V
2.5V RGMII/GMII	Supply Voltage 2.5V	-0.2	2.8	V
1.8V RGMII	Supply Voltage 1.8V	-0.2	2.3	V
1.5V RGMII	Supply Voltage 1.5V	-0.2	2.0	V
3.3V DCinput 3.3V DCoutput	Input Voltage Output Voltage	-0.3	3.6	V
1.0V DCinput 1.0V DCoutput	Input Voltage Output Voltage	-0.3	1.2	V
NA	Storage Temperature	-55	+125	°C

Note: Refer to the most updated schematic circuit for correct configuration.

10.2. Recommended Operating Conditions

Table 53. Recommended Operating Conditions

Description	Pins	Minimum	Typical	Maximum	Unit
Supply Voltage VDD	DVDD33, AVDD33	2.97	3.3	3.63	V
	AVDD10, DVDD10	0.95	1.0	1.05	V
	2.5V RGMII/GMII	2.25	2.5	2.75	V
	1.8V RGMII	1.62	1.8	1.98	V
	1.5V RGMII	1.5	1.55	1.6	V
Ambient Operating Temperature T _A (RTL8211F/RTL8211FD)	-	0	-	70	°C
Ambient Operating Temperature T _A (RTL8211FI/RTL8211FDI)	-	-40	-	85	°C
Maximum Junction Temperature	-	-	-	125	°C

10.3. Crystal Requirements

Table 54. Crystal Requirements

Symbol	Description/Condition	Minimum	Typical	Maximum	Unit
F _{ref}	Parallel Resonant Crystal Reference Frequency, Fundamental Mode, AT-Cut Type.	-	25	-	MHz
F _{ref} Tolerance	Parallel Resonant Crystal Frequency Tolerance, Fundamental Mode, AT-Cut Type. T _a =0°C~70°C.	-50	-	+50	ppm
F _{ref} Duty Cycle	Reference Clock Input Duty Cycle.	40	-	60	%
ESR	Equivalent Series Resistance.	-	-	50	Ω
DL	Drive Level.	-	-	0.5	mW
Jitter	Broadband Peak-to-Peak Jitter ^{1, 2}	-	-	200	ps
V _{ih} _CKXTAL	Crystal Output High Level	1.4	-	-	V
V _{il} _CKXTAL	Crystal Output Low Level	-	-	0.4	V

Note 1: 25kHz to 25MHz RMS < 3ps.

Note 2: Broadband RMS < 9ps.

Note3: F_{ref} Tolerance +/- 50ppm including effects of aging of the first year, external crystal capacitors, and PCB layout.

10.4. Oscillator/External Clock Requirements

Table 55. Oscillator/External Clock Requirements

Parameter	Condition	Minimum	Typical	Maximum	Unit
Frequency	-	-	25/50	-	MHz
Frequency Tolerance (RTL8211F/RTL8211FD)	T _a =0°C~70°C	-50	-	50	ppm
Frequency Tolerance (RTL8211FI/RTL8211FDI)	T _a =-40°C~85°C	-50	-	50	ppm
Duty Cycle	-	40	-	60	%
Broadband Peak-to-Peak Jitter ^{1, 2}	-	-	-	200	ps
V _{peak-to-peak}	-	3.15	3.3	3.45	V
Rise Time (10%~90%)	-	-	-	10	ns
Fall Time (10%~90%)	-	-	-	10	ns
Operating Temperature Range	-	-40	-	85	°C

Note 1: 25kHz to 25MHz RMS < 3ps.

Note 2: Broadband RMS < 9ps.

Note3: Frequency Tolerance +/- 50ppm including effects of aging of the first year, external crystal capacitors, and PCB layout.

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10.5. DC Characteristics

Table 56. DC Characteristics

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
VDD33, AVDD33	3.3V Supply Voltage	-	2.97	3.3	3.63	V
1. MDIO (Table 4, page 8) 2. RGMII I/O (Table 3, page 8)	2.5V RGMII Supply Voltage	-	2.25	2.5	2.75	V
1. MDIO (Table 4, page 8) 2. RGMII I/O (Table 3, page 8)	1.8V RGMII Supply Voltage	-	1.62V	1.8V	1.98V	V
1. MDIO (Table 4, page 8) 2. RGMII I/O (Table 3, page 8)	1.5V RGMII Supply Voltage	-	1.5V	1.55V	1.6V	V
DVDD10, AVDD10	1.0V Supply Voltage	-	0.95	1.0	1.05	V
Voh (3.3V)	Minimum High Level Output Voltage	-	2.4	-	VDD33 + 0.3	V
Voh (2.5V)	Minimum High Level Output Voltage	-	2.0	-	VDD25 + 0.3	V
Voh (1.8V)	Minimum High Level Output Voltage	-	0.9*VDD18	-	VDD18 + 0.3	V
Voh (1.5V)	Minimum High Level Output Voltage	-	0.9*VDD15	-	VDD15 + 0.3	V
Vol (3.3V)	Maximum Low Level Output Voltage	-	-0.3	-	0.4	V
Vol (2.5V)	Maximum Low Level Output Voltage	-	-0.3	-	0.4	V
Vol (1.8V)	Maximum Low Level Output Voltage	-	-0.3	-	0.1*VDD18	V
Vol (1.5V)	Maximum Low Level Output Voltage	-	-0.3	-	0.1*VDD15	V
Vih (3.3V)	Minimum High Level Input Voltage	-	2.0	-	-	V
Vil (3.3V)	Maximum Low Level Input Voltage	-	-	-	0.8	V
Vih (2.5V)	Minimum High Level Input Voltage	-	1.7	-	-	V
Vil (2.5V)	Maximum Low Level Input Voltage	-	-	-	0.7	V
Vih (1.8V)	Minimum High Level Input Voltage	-	1.2	-	-	V
Vil (1.8V)	Maximum Low Level Input Voltage	-	-	-	0.5	V
Vih (1.5V)	Minimum High Level Input Voltage	-	1.0	-	-	V
Vil (1.5V)	Maximum Low Level Input Voltage	-	-	-	0.3	V
Iin	Input Current	Vin=VDD33 or GND	0	-	0.5	μA

Note: Pins not mentioned above remain at 3.3V.

10.6. AC Characteristics

10.6.1. MDC/MDIO Timing

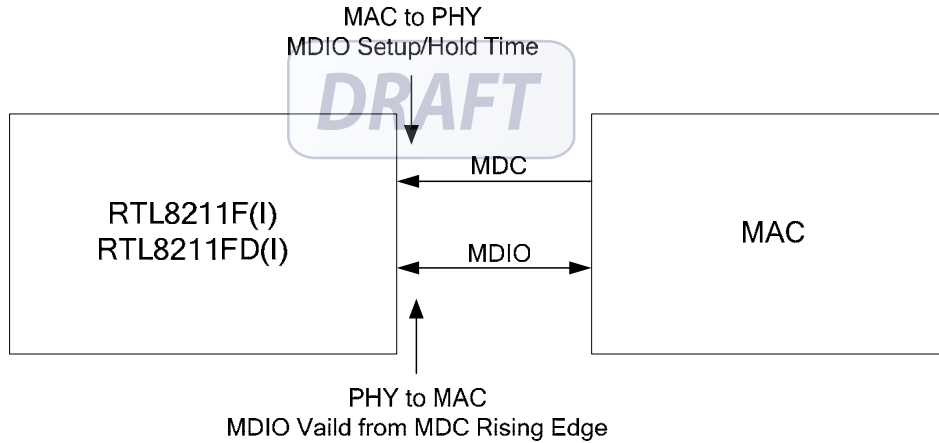


Figure 25. MDC/MDIO Setup, Hold Time, and Valid from MDC Rising Edge Time Definitions

MDC/MDIO Timing – Management Port

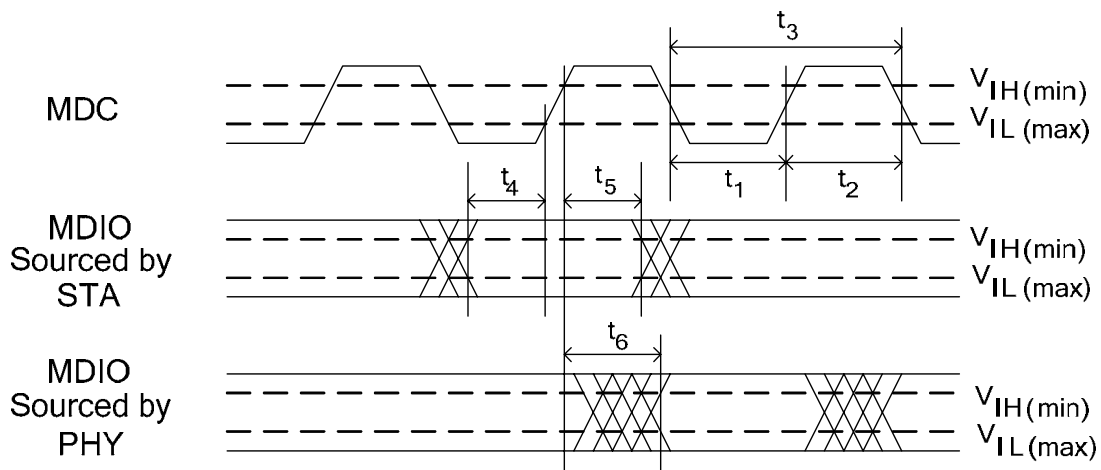


Figure 26. MDC/MDIO Management Timing Parameters

Table 57. MDC/MDIO Management Timing Parameters

Symbol	Description	Minimum	Maximum	Unit
t_1	MDC High Pulse Width	32	-	ns
t_2	MDC Low Pulse Width	32	-	ns
t_3	MDC Period	80	-	ns
t_4	MDIO Setup to MDC Rising Edge	10	-	ns
t_5	MDIO Hold Time from MDC Rising Edge	10	-	ns
t_6	MDIO Valid from MDC Rising Edge	0	300	ns

10.6.2. RGMII Timing Modes

Timing for this interface will be such that the clock and data are generated simultaneously by the source of the signals and therefore skew between the clock and data is critical to proper operation.

Figure 27 shows the effect of adding an internal delay to TXC when in RGMII mode.

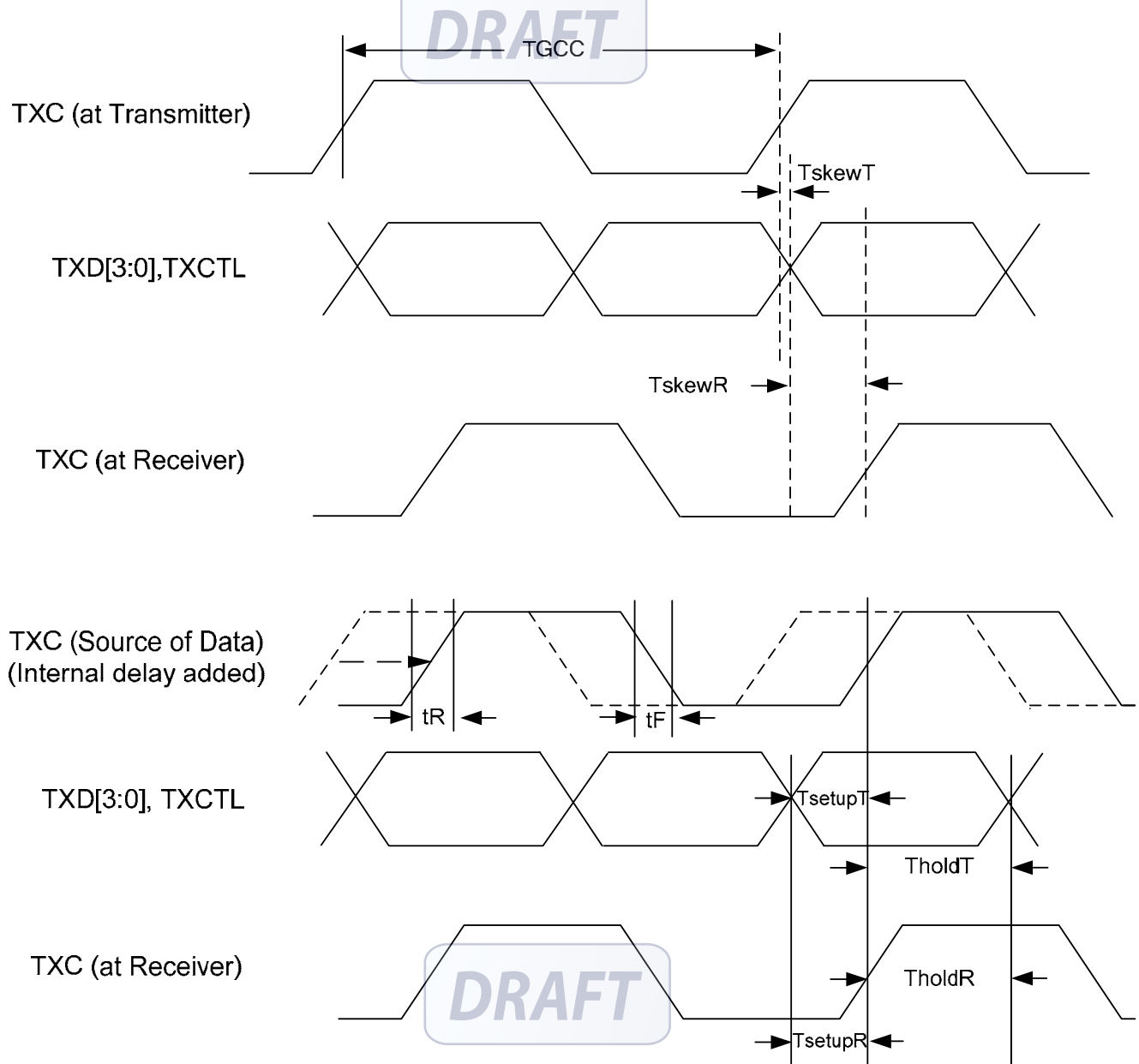


Figure 27. RGMII Timing Modes (For TXC)

Figure 28 shows the effect of adding an internal delay to the RXC flow when in RGMII mode.

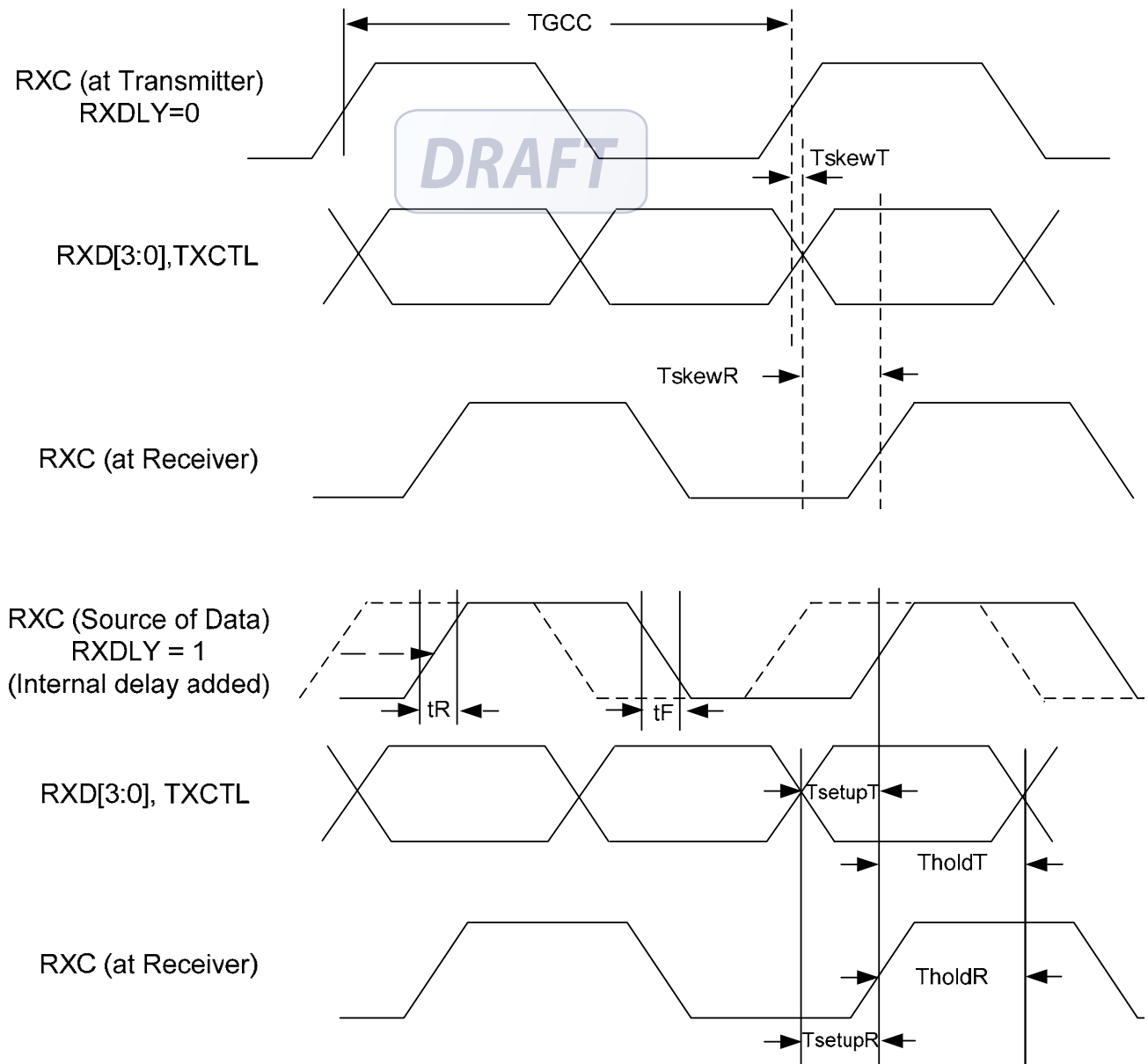


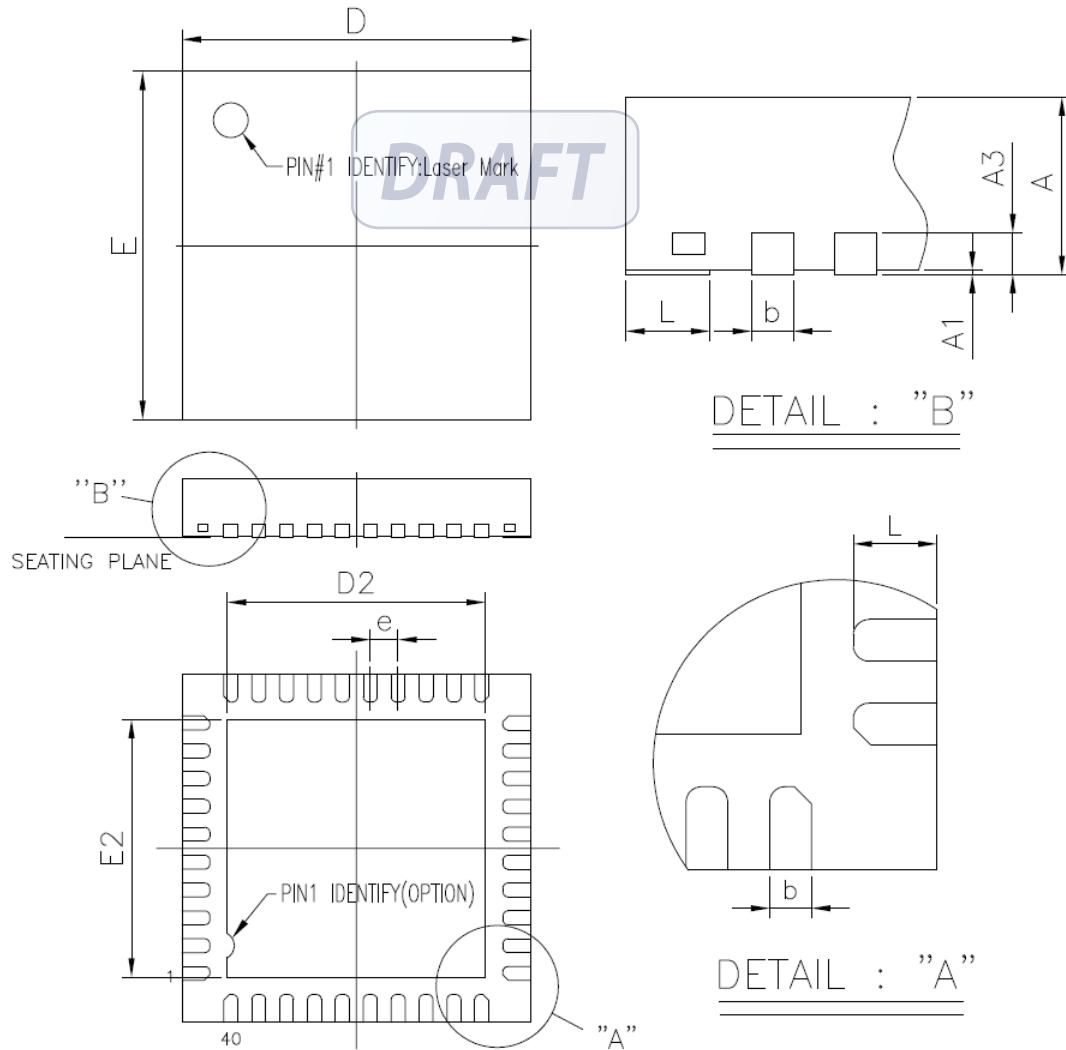
Figure 28. RGMII Timing Modes (For RXC)

Table 58. RGMII Timing Parameters

Symbol	Description	Min	Typical	Max	Units
TGCC	Clock Cycle Duration (1000Mbps)	7.2	8	8.8	ns
	Clock Cycle Duration (100Mbps)	36	40	44	ns
	Clock Cycle Duration (10Mbps)	360	400	440	ns
Duty_G	Duty Cycle for 1000	45	50	55	%
Duty_T	Duty Cycle for 10/100	40	50	60	%
tR	TXC/RXC Rise Time (20%~80%)	-	-	0.75	ns
tF	TXC/RXC Fall Time (20%~80%)	-	-	0.75	ns
TsetupT	Data to Clock Output Setup (at transmitter integrated delay)	1.2	2	-	ns
TholdT	Data to Clock Output Hold (at transmitter integrated delay)	1.2	2	-	ns
TsetupR	Data to Clock Input Setup (at receiver integrated delay)	1.0	2	-	ns
TholdR	Data to Clock Input Hold (at receiver integrated delay)	1.0	2	-	ns
TskewT	Data to Clock Output Skew (at transmitter)	-0.5	0	0.5	ns
TskewR	Data to Clock Input Skew (at receiver) This implies that PC board design will require clocks to be routed such that an additional trace delay of greater than 1.5ns and less than 2.0ns will be added to the associated clock signal.	1	1.8	2.6	ns

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11. Mechanical Dimensions



11.1. Mechanical Dimensions Notes

Symbol	Dimension in mm			Dimension in inch		
	Min	Nom	Max	Min	Nom	Max
A	0.80	0.85	0.90	0.031	0.033	0.035
A ₁	0.00	0.02	0.05	0.000	0.001	0.002
A ₃	0.20REF			0.008REF		
b	0.15	0.20	0.25	0.006	0.008	0.010
D/E	5.00 BSC			0.197 BSC		
D2/E2	3.45	3.70	3.95	0.136	0.146	0.156
e	0.40 BSC			0.016 BSC		
L	0.30	0.40	0.50	0.012	0.016	0.020

Note 1: CONTROLLING DIMENSION: MILLIMETER (mm).

Note 2: REFERENCE DOCUMENT: JEDEC MO-220.

12. Ordering Information

Table 59. Ordering Information

Part Number	Package	Status
RTL8211F-CG	40-Pin QFN with 'Green' Package (Switching Regulator model)	ES ready MP in Mar/2014
RTL8211FD-CG	40-Pin QFN with 'Green' Package (LDO model)	ES ready MP in Mar/2014
RTL8211FI-CG	40-Pin QFN with 'Green' Package (Switching Regulator model) Industrial grade.	-
RTL8211FDI-CG	40-Pin QFN with 'Green' Package (LDO model). Industrial grade.	-

Note: See page 6 for package identification.

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